kayfabe

A Mode-2 NVIDIA GPU forwarding stack for KVM/QEMU guests

An architecture whitepaper, written to be attacked

Subject. Two repositories on one machine. `/workspace/nvidia-gpu-passthrough` — the C research artifact, 739 commits, 44 031 lines of C, developed 2026-05-24 to 2026-07-24; it works on real hardware. `/workspace/nvkvm-rs` — **kayfabe**, a clean-slate Rust rewrite, 107 commits, 42 320 lines of implementation and 43 272 lines of test code, developed 2026-07-24 to 2026-07-28; it has never touched a GPU. This document is pinned to Rust HEAD `6ce8599` and C HEAD `2899a74`.

What moved since the previous pin (`0b78102`), and why the pin was updated rather than the text patched. Fourteen commits landed in between. One of them, `f2055bf`, built `kayfabe-gsp` — the crate this paper's previous revision named as its headline risk and described as "34 lines". It is now 3 550 lines across 8 modules. The GSP sections and two of the five figures were rewritten rather than decremented, because the *shape* of the risk changed and not only its size (§8.2). Three other revisions are folded in: the L2-Q QEMU-adapter design (`b31487a`, §8.7), the 580-vs-610 driver-version correction (`7af23cb`), and the ring gate becoming structural rather than a call-graph property (`634b253`). A pinned document that has been edited past its pin is worse than an unpinned one; every number below was re-measured at `6ce8599`.

Stance. Every number here was measured from the tree while writing, not copied from a document. Where a project document and the code disagree, the code wins and the disagreement is recorded — there is a list of them in §10, and it is one of the more useful parts of this paper. Roughly a third of the document is about what does not work, is not built, or is not known.

Contents

1 What this document is, and how to attack it

This is a review document. Its job is to describe kayfabe accurately enough that a competent systems engineer who has never seen it can (a) explain what it does and why it is hard, (b) find any component in the tree, (c) judge whether the claims are believable, and (d) find the weak points without help. It is not a status report and it is not a pitch.

Four consequences shape everything below.

Numbers trace to something. Every count, score and measurement names its source. Where a number is an estimate it says so; where a claim could not be verified it is marked [unverified] rather than softened. Appendix A gives the exact commands used.

Unbuilt is stated as unbuilt. Most of the planned system does not exist. The build order is deliberate, but a document that blurs "designed" and "running" is worthless for the stated purpose. The layer diagram (Figure 1) and the data-path diagram (Figure 3) both carry explicit **NOT BUILT** bands.

The uncomfortable section is the longest. §8 is where to start if you only read one section.

This document applies the project's own doctrine to itself. kayfabe's testing doctrine opens with the rule *"a green instrument on an unexercised path is worse than no instrument — it reads as evidence, and nobody re-checks a zero"* (`docs/design/testing_doctrine.md` §1). The same rule applies to a whitepaper. So where this document says a thing is tested, it also says what that test would have to be wrong about for the claim to fail.

The single most important sentence in this paper. No part of the Rust codebase has ever driven a real GPU. There is no binary in the workspace: `find -name main.rs` returns exactly one hit, and it is the offline ABI generator, which lives outside the workspace on purpose. The only composition of vCPU threads, reactor, executor and core that exists is in the test suite (`tests/tests/l1_mean.rs`, `guest_mean_run`). **This is still true at this revision**, and it is true of the newly-built faked GSP too: `kayfabe-gsp` decodes real `msgq` elements and real RPC envelopes, but only ones a test wrote into a `BTreeMap`. Everything below describes a system that is real, tested and internally coherent, and that has not yet met the thing it exists to talk to.

2 The idea

2.1 The problem

Giving a virtual machine a real GPU normally means one of two things. **IOMMU passthrough** hands the whole physical device to exactly one guest: simple, fast, and it gives every other guest nothing. **Vendor virtualization** (SR-IOV, NVIDIA vGPU) shares the device properly, but it exists on datacenter parts under licences, not on the consumer cards most people own.

kayfabe is a third option: **share one commodity GPU among several untrusted virtual machines, with per-guest-process blast-radius containment, using only unprivileged host processes**. The multi-tenant part is the point. A single cross-tenant leak, or a hostile guest that can wedge the box, would be fatal to the thesis.

2.2 Vocabulary

The rest of this paper leans on five terms.

- **RM** — NVIDIA's *Resource Manager*, the object model inside the driver. Everything a GPU program touches is an RM object with a handle: clients, devices, address spaces, channels, memory, engine contexts. Guest software allocates and frees them through ioctls and RPCs.
- **GSP** — the *GPU System Processor*: on modern NVIDIA hardware a real CPU core on the GPU die, running a firmware copy of much of the resource manager. The host driver no longer programs most of the hardware directly; it sends RPC messages to the GSP over a ring buffer in memory.
- **VAS / PDB** — a GPU virtual address space, and its *page directory base*: the physical address of the root of that address space's page tables. The GPU's MMU keys page tables by PDB, which makes **the PDB the real memory boundary** — not the client handle.
- **Channel / vChid** — channels are the GPU's work-submission queues; the *virtual channel id* is how a submission is routed to one. **vChid is the execution boundary.**
- **Doorbell** — the MMIO register write that says "channel N has work waiting". The value written is a token that encodes the vChid.

2.3 Mode 1 and Mode 2

The predecessor project defines two modes, and kayfabe is only the second.

Mode 1 forwards the guest's *userspace* NVIDIA ioctls to a host stub that replays them against the real `/dev/nvidia*`. It needs a guest kernel module and a virtio transport, so the guest is modified, and it caps you at "a Linux guest running our module". It is mature: 22 real GPU applications at host parity, Vulkan, OpenGL, a working desktop present path, five security audits.

Mode 2 inverts it. The guest runs the **stock, unmodified NVIDIA kernel driver** against an *emulated* PCI device that presents a plausible GPU and a *faked GSP*. We implement the RPC ring the driver talks to and answer as the firmware would. The guest driver believes it owns hardware: it allocates RM objects, builds GPU page tables, creates channels, and rings doorbells. We are on the other side of that conversation, and from the protocol traffic we reconstruct what the guest program was actually trying to do — then do the equivalent thing on a real host GPU through ordinary unprivileged userspace operations, inside a sandbox.

Mode 2 needs *nothing* from the guest, which is the entire reason it is worth the difficulty.

2.4 The correctness criterion, and why it is not "replay everything"

The naive reading of Mode 2 is "intercept the guest's privileged operations and perform them on the host". That is wrong, and the project's forwarding model says so explicitly:

> *"The job of Mode-2 is **not** to replay those low-level steps on the host. It is to **recover the original userspace-level intent and re-express it as the normal, unprivileged host userspace operations** — exactly the operations Mode-1 forwards directly."*
> [C: docs/design/mode2_forwarding_model.md]

The guest's kernel RM has *already* decomposed an unprivileged userspace intent (`cuCtxCreate`, `cuMemAlloc`, a kernel launch) into privileged GSP-internal steps. Replaying those steps on the host would require privilege we deliberately do not have — and the project has a named lesson about this: an `NV_ERR_INSUFFICIENT_PERMISSIONS` from the host stub *never* means "we lack a privilege we should have", it means "we forwarded at the wrong layer". So the operational split is:

- **Case 1** — the RPC essentially *is* the userspace operation. Re-issue it roughly 1:1 on the host, through the isolate, unprivileged, and let the host's own kernel RM legitimately re-derive the privileged steps for itself.
- **Case 2** — a GSP-internal control with no userspace equivalent (`PROMOTE_CTX`, `GET_CTX_BUFFER_INFO`). **Acknowledge the guest; do nothing on the host.** The host has already done the equivalent for its own context.

The correctness criterion that licenses this is **observable end-state equivalence**, not step fidelity. It is perfectly correct for some guest-kernel operations to complete instantly as fakes, *as long as* the one operation that actually carries the work triggers the real host-side chain. The hard boundary on the other side is equally explicit: a completion that gates guest *userspace* must be a real host write. Forging one would make the whole project a lie, and the design forbids it by name.

2.5 Why this is hard

1. **The protocol is not documented and not stable.** It varies on two independent axes: the *driver version* (wire layouts) and the *GPU generation* (silicon behaviour). kayfabe separates these as "Axis A" and "Axis B" and gives each its own trait seam, because conflating them is its own bug class.
2. **Some things cannot be faked at all.** GR's golden context is produced by FECS/GPCCS microcode on real silicon. There is no way to fabricate one. The design's response is to never emulate an engine — forward the guest's own pushbuffer and let real silicon produce the context.
3. **Address translation is a trust problem, not a lookup problem.** The guest builds GPU page tables in memory it controls. Reading them at an arbitrary instant means acting on bytes an adversary can change under you.
4. **Identity collides by construction.** Guest handles and guest VAs are per-process namespaces. Two processes using identical values is normal. Anything that keys on them is wrong (Figure 4).
5. **The failure modes are global.** A mis-resolved address is not a wrong answer; it is a host GPU MMU fault that takes out a channel, and in the worst case a state the guest driver cannot recover from without a full VM restart.
6. **Everything is concurrent.** N guest vCPUs, several guest processes, several host isolate processes, an interrupt path, and a host RM that *serializes every ioctl per client* so that one wedged verb blocks every

isolate in D-state.

3 Provenance: the C artifact, and what it actually proved

kayfabe is a rewrite, not a greenfield project. The C artifact at `/workspace/nvidia-gpu-passthrough` is the working prior implementation and the differential oracle for everything the rewrite claims about NVIDIA's behaviour. Being precise about what it did and did not prove is the foundation of every other claim here.

Claim	Mode	What was actually measured
22 real GPU apps at host parity	1	**True.** `tests/perf/realapp_matrix.md`, 2026-06-01. Same binary, host and guest, strictly serially on one RTX 3060, with a correctness check alongside throughput; parity gate ≥ 0.90. 10 CUDA benchmarks, 7 PyTorch, 2 LLM, 3 graphics. **Caveat the doc states itself:** NVENC is *partial*, not parity (`InitializeEncoder` fails); NVDEC excluded (broken on the host baseline too).
7B LLM at 63 tok/s	1	**True but mis-framed.** 63.4 tok/s is a Mode-1 A/B endpoint after a caching fix, not a headline result. The audited same-day parity figure is **host 67.8 → guest 66.0 tok/s, 0.97×** (Qwen2.5-7B Q4_K_M, `-ngl 99`, RTX 3060). Quote the ratio.
Mode-2 bare-metal parity	2	**True, one workload.** 49.9 tok/s Mode-2 vs 47.5 tok/s host-native *on the same RTX 3050*, i.e. zero forwarding overhead within noise. The entire earlier 20→50 tok/s gap was **nested-virt vmexit tax**, not Mode-2 design — the vast.ai bench is itself a VM, so the Mode-2 guest was L2. The project retracted its own earlier "~21% residual" claim as an apples-to-oranges comparison across two different GPUs.
Mode-2 CUDA correctness	2	**True.** Full fake-boot + GSP RPC + GMMU + CE + IRQ; `cuInit`, `cuCtxCreate`, byte-exact matmul; PyTorch 2.5.1 *single-process*. Bugs #12 (second-context hang) and #13 (multi-iteration hang) resolved on hardware.
What the C artifact never proved		
Mode-2 multi-process	2	**Never.** Bug #14 is open. Root-caused on hardware 2026-07-24 by a host Xid 31 `FAULT_PDE` naming the loser's exact GR channel: the second process's identical guest VAs were never published into *its own* host address space.
Mode-2 sequential processes	2	**Broken at HEAD.** Measured 2026-07-25: exactly *one* CUDA process works per QEMU lifetime; the second fails `cuInit`→999 regardless of how the first exited. The "fresh boot per clean run" bench discipline *is* this bug, not a precaution.
Mode-2 graphics / display	2	**Never.** All Vulkan/OpenGL/present results are Mode 1.
Mode-2 on the 22-app matrix	2	**Never run.** It is Mode-2's definition of done.
Multi-GPU, MIG, non-Ampere	—	**Never.** Ampere GA106 only. Turing/Ada/Hopper/Blackwell rows in the ABI doc are all marked *assumption — verify*.
Mode-2 security posture	2	**Not audited.** The five security audits are Mode-1-era. The 2026-07-25 bench found live guest-reachable Mode-2 defects, including parsing arbitrary guest RAM as GSP RPC elements after a guest-triggered driver reload.

The structural reason for the rewrite. #14 is not a bug that could be patched. The architecture document states it in one line: *"the emulator's completion, execution, isolate, and GPA-arena planes are each a single-process singleton, and they fail in dependency order."* Four rounds of investigation each found a different plane; they were one structural fact observed four times. Making per-process separation structural rather than retrofitted is the rewrite's founding requirement, and it is why the diagram in Figure 4 is the centre of the design.

One provenance correction worth carrying. Two C design documents state the C baseline encodes *"~14 months of quirk capture"*. Git says the first commit is 2026-05-24 and the last is 2026-07-24: **two months, 739 commits**. Either the figure means agent-hours, or it is wrong. Use the verifiable number.

4 The architecture

4.1 Layers and ports

The one defining constraint. The logic core is a pure state machine over guest-supplied bytes: no OS, no syscalls, no hypervisor types, no wall clock, no `#[repr(C)]` NVIDIA struct layouts, and no concrete GPU-generation or driver-version name anywhere. Everything effectful crosses a trait seam. `unsafe_code = "forbid"` is a workspace lint; every core type is compile-time-asserted `Send + Sync`.

This is enforced, not intended. Four CI greps run on every push: a *hexagonal boundary* grep that fails if `eventfd|epoll|timerfd|rawfd|libc|O_NONBLOCK` appears in any of the 11 pure crates; a *VMM-vocabulary* grep that fails if any QEMU concept (`BQL`, `MemoryRegion`, `qemu_bh`, `iothread`, …) appears in those 11 plus `kayfabe-rt`; a *generation-name* grep over 10 of them that fails on any concrete chip or GPU-generation name outside a comment; and a *GPA-accessor* gate that asserts `.gpa_read(`/`.gpa_write(` appears **exactly once** in the entire forwarding crate, and **nowhere** in the other nine.

The claim that pays for this constraint is the **adapter contract**: bringing up a new GPU generation is `impl Arch for <Gen>` in an adapter crate with *zero edits to any logic crate*. The standing proof is that the whole suite runs the real core against `MockArch`, whose encodings are deliberately non-NVIDIA — if the core had absorbed a real encoding anywhere, the mock would not work.

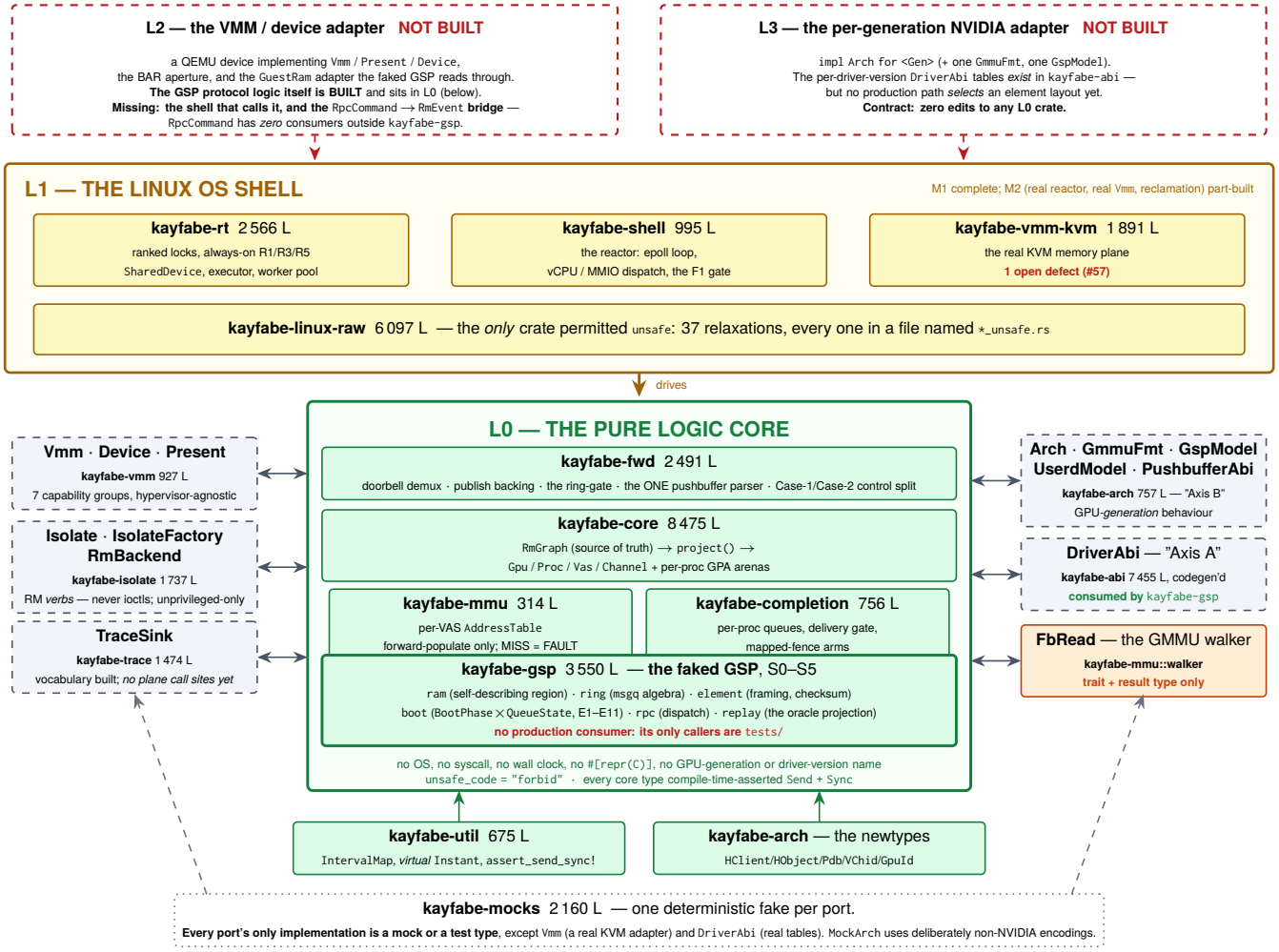


Figure 1: The layer model and the hexagonal ports. Line counts are measured at `6ce8599` (`crates/*/src/**/*.rs`; `kayfabe-abi`'s figure includes its detached offline generator). Implementor census, counted from `impl <Trait> for` across the tree: `FbRead` has *no* implementation of any kind; `Device`, `Present`, `Arch`, `GspModel`, `Isolate`, `IsolateFactory` and `RmBackend` have only mocks or test types; `Vmm` has one real adapter (`KvmVmm`); `TraceSink` has two in-crate implementations. `kayfabe-gsp` joined the L0 band at this revision — it is a pure logic crate and is covered by all four CI boundary greps — but the L2 shell that would call it does not exist.

4.2 The crate dependency graph

Figure 2 is derived from every `Cargo.toml` in the tree and cross-checked against `Cargo.lock`; it is not drawn from the design documents. Three findings come straight off it.

The external dependency surface is essentially zero. In the whole workspace there are three external crates: `libc` — a real dependency of `kayfabe-linux-raw` only, and a *dev*-dependency of `kayfabe-vmm-kvm` and `tests` for asserting exact errnos — plus `trybuild` and `proptest`, both dev-only. Thirteen of the sixteen crates have no external dependency of any kind. Whatever else is true, the supply-chain surface is not a place to look for problems.

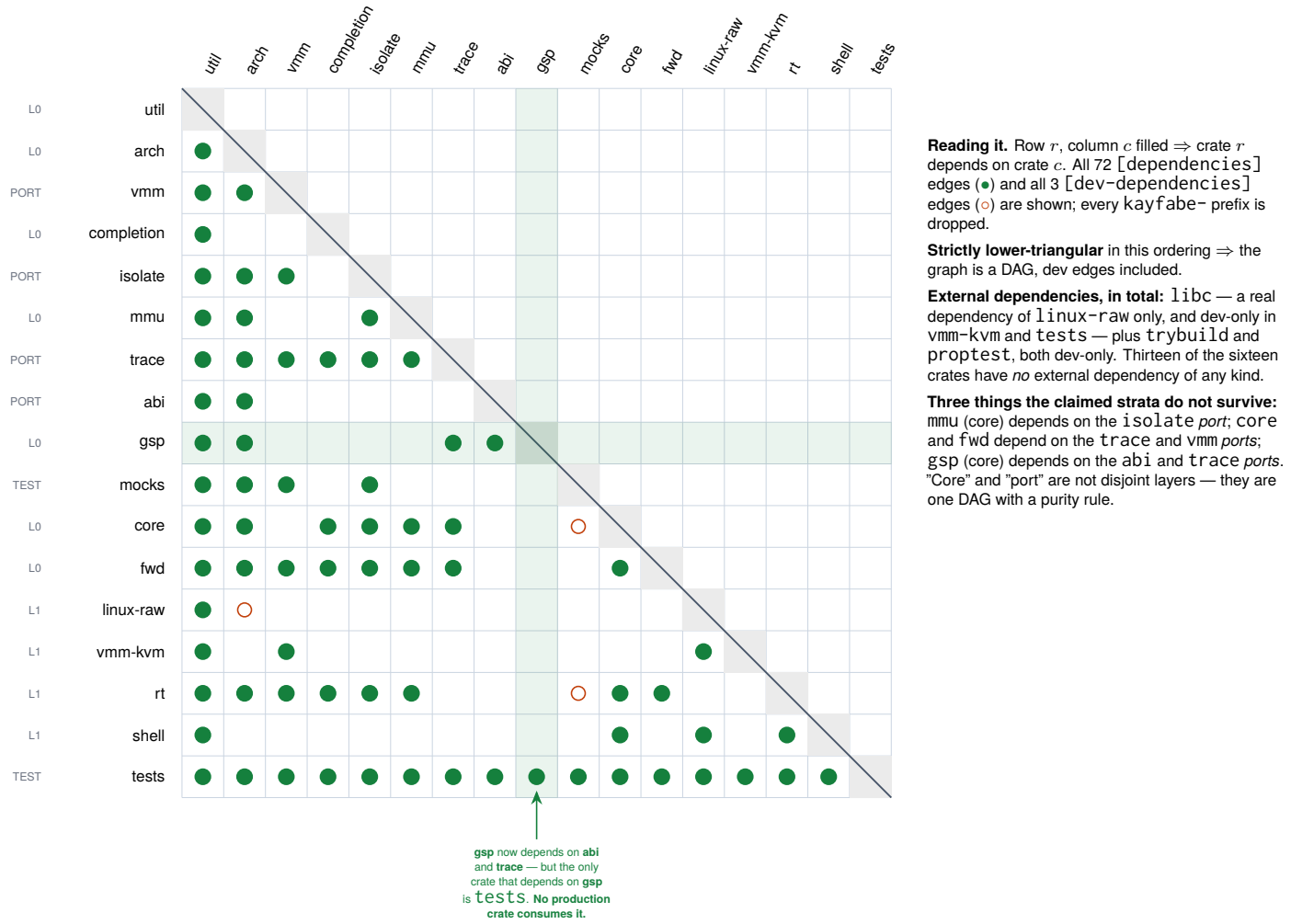
The graph is acyclic, including dev edges, which the strictly lower-triangular shape of the matrix demonstrates directly. *The ordering that makes it lower-triangular had to change at this revision:* `gsp` was a leaf sitting above `abi` and `trace`, and now depends on both, so it moved below them. The re-ordering is the matrix's own record of the crate ceasing to be a leaf.

But the claimed strata are not disjoint, and the documents imply they are. `kayfabe-mmu` — a core crate — depends on the `kayfabe-isolate` *port* (it needs `HostHandle`). `kayfabe-core` and `kayfabe-fwd` depend on the `trace` and `vmm` ports. `kayfabe-gsp` — also a core crate — depends on the `abi` and `trace` ports. "Core" and "port" are one DAG with a purity rule, not two layers. That is a defensible design (the trace crate had to sit *below* core so every plane could emit into it), but a reader who takes the hexagon picture literally will be surprised by the manifests.

The previous revision of this paper called `kayfabe-abi` and `kayfabe-gsp` "disconnected islands: nothing in the workspace — not `tests`, not `fuzz` — depends on either". That was true then. Half of it is false now, and the surviving half is sharper.

Re-checked against the manifests and the source, not inherited. `kayfabe-abi` is no longer an island: `crates/kayfabe-gsp/Cargo.toml` lists it, and five of the eight `kayfabe-gsp` modules use it (`lib`, `boot`, `element`, `fault`, `replay`) — it supplies the RPC envelope view and the version-keyed layout values. So the 7 455 lines of oracle-tested wire decoding finally have a consumer, and that consumer is the one the crate was written for.

`kayfabe-gsp` **is still an island in the direction that matters**: the only manifest that lists it is `tests/Cargo.toml`. No production crate depends on it, nothing constructs its inputs outside `tests/`, and its output type `RpcCommand` has **zero** references anywhere in the tree outside the crate that defines it. It is 3 550 lines of working protocol logic with a test suite and no caller — which is exactly as interesting as the old caption said, one level up the stack.



Reading it. Row r, column c filled ⇒ crate r depends on crate c. All 72 `[dependencies]` edges (●) and all 3 `[dev-dependencies]` edges (○) are shown; every `kayfabe-` prefix is dropped.

Strictly lower-triangular in this ordering ⇒ the graph is a DAG, dev edges included.

External dependencies, in total: `libc` — a real dependency of `linux-raw` only, and dev-only in `vmm-kvm` and `tests` — plus `trybuild` and `proptest`, both dev-only. Thirteen of the sixteen crates have *no* external dependency of any kind.

Three things the claimed strata do not survive: `mmu` (core) depends on the `isolate` *port*; `core` and `fwd` depend on the `trace` and `vmm` *ports*; `gsp` (core) depends on the `abi` and `trace` *ports*. ”Core” and ”port” are not disjoint layers — they are one DAG with a purity rule.

Figure 2: The crate dependency matrix, derived from the manifests at `6ce8599`. `kayfabe-abi` is no longer an island — `kayfabe-gsp` consumes it, which is what it was written for. `kayfabe-gsp` still is one: the only manifest that names it is `tests/Cargo.toml`, so 3 550 lines of protocol logic have a test suite and *no production caller*. Note also that the previous revision’s matrix omitted one real edge, `tests` → `rt`; it showed 65 dependency edges where the manifests had 66.

4.3 The data path

Figure 3 traces one guest GPU operation from a CUDA call to real silicon and back, naming the symbol each step lives at. Three things about it are worth stating in prose.

The plan/execute/commit shape is the whole concurrency design. A host RM verb can block for a long time — the host RM serializes every ioctl per client, and it waits *uninterruptibly*, so one wedged verb parks every isolate that shares the client in D-state. Invariant **R1** therefore forbids issuing a verb while any ranked lock is held. That forces a three-phase shape: *plan* under locks, *execute* with no lock held, *commit* after re-acquiring. And because the locks were dropped in the middle, invariant **R5** requires the commit to re-validate everything the plan assumed. This is not free — §8 covers what it costs.

The ring gate is the #14 fix, and it is now structural in the type system. Before any host operation, a doorbell submission's entire working set must already be published into *this* `Vas`'s own host address space. If it is not, the call refuses with `NoVas` having issued zero host operations. The failure it prevents is exactly the one the C artifact hit on real hardware: the losing process's identical guest VAs were never published into its own host GR address space, so the host GPU faulted past the shared prefix. At the previous pin this was a call-graph property — "every path happens to go through `plan_doorbell`". At `634b253` it stopped being one: `VerbPlan::Doorbell` is `#[non_exhaustive]`, so no struct expression for it exists outside `kayfabe-isolate`, and the sole constructor `VerbPlan::gated_doorbell` *runs* the gate. Constructing an ungated doorbell is now compile error E0639, pinned by a `trybuild` row (`crates/kayfabe-isolate/tests/ui/`).

The upper half of the diagram is half-built, and the half that is missing moved. At the previous pin there was no register model, no GSP message-queue transport and no RPC decode. All three now exist, in `kayfabe-gsp`. What still does not exist is (a) any adapter that implements `GuestRam` or `Device` outside `tests/`, and (b) the bridge from the GSP's decoded output to the core's input — `RpcCommand` has zero references anywhere in the tree outside the crate that defines it. So the headline sentence survives verbatim: **nothing in the repository produces an `RmEvent` from real guest bytes**, and the core's control-plane input is still synthesised by the test suite. §8.2 is about what changed underneath that sentence.

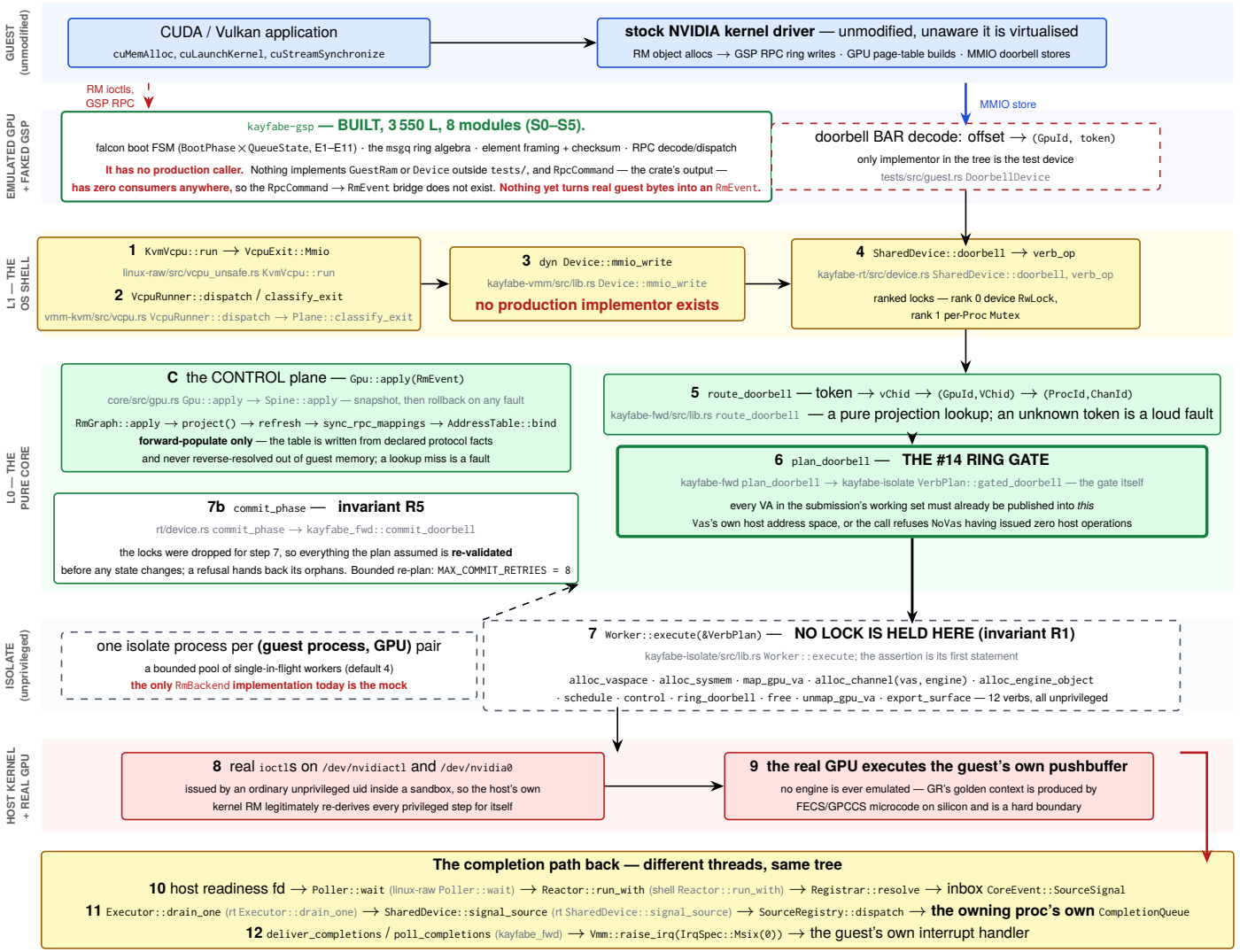


Figure 3: The data path, verified function by function against the tree at `6ce8599`. Numbered steps are the execution plane; **C** is the control plane, which populates the address table from declared protocol facts. Boxes name *symbols*, not line numbers: the repository converted its own 105 pins to symbols in `3fd1455` after they drifted, and six of this figure's pins had drifted by this revision. Red text marks what does not exist.

4.4 The isolation model

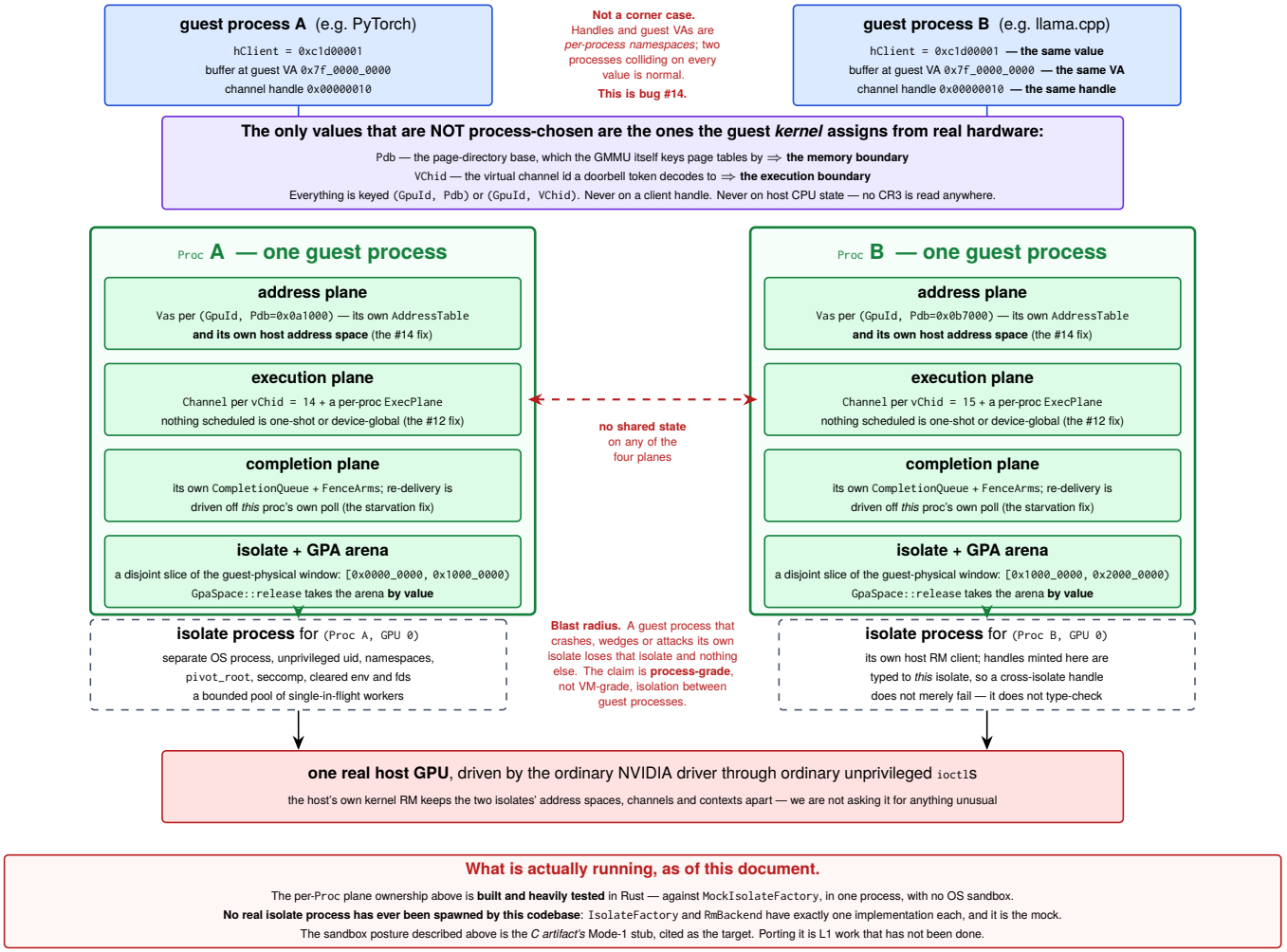


Figure 4: The process and isolation model, and the problem it exists to solve. The per-Proc ownership of all four planes is built and heavily tested against mocks; the sandboxed isolate process at the bottom has never been spawned by this codebase.

The rule that generates the whole model is one sentence: **a guest process is a grouping label, and it is never what an address or execution operation keys on.** Address operations key on `(GpuId, Pdb)`; execution operations key on `(GpuId, VChid)`. `Proc` exists only to own the isolate, the GPA arena and the lifecycle.

`Proc` membership is itself a **pure projection** of the RM graph — the dup-connected components of declared user clients — never inferred from timing, never from arrival order. That matters because it means two identical event streams delivered in different orders produce bit-identical process boundaries, which is a property the test suite checks directly by permuting streams.

4.5 The lifecycle

The law every path obeys: RETIRE EAGER, REAP DEFERRED

Retire is immediate and cheap: the `Proc` stops accepting operations, its isolates refuse new check-outs. **Reap** is the heavy half — freeing host objects, killing isolate processes, recycling the GPA arena — and it runs later, at a quiesce point the adapter declares (`Gpu::reap_retired`).

Why: the C artifact proved that reaping at the client-root free *hangs the dying context's own residual polls*. Reaping never is the #80 leak. Both were paid for on hardware.

TRIGGER	WHAT HAPPENS	STATUS — read from the tree
T0 a guest frees a subset and keeps running	The most frequent path in a real workload, and **the one with no backstop** — nothing dies, so nothing reclaims. Dropped host identities move to a `pending_release` queue on the `Proc` and drain on a checked-out worker.	**Built, and lazier than designed.** The drain may fire only when the isolate is otherwise idle (`is_quiesced`), because no lock can exclude a verb that runs lock-free.
T1 a verb is cancelled mid-flight	The cancel is latched, discharged with no lock held, surfaces as `EINTR` → `Interrupted` → `FwdFault::Cancelled`; the chain's own unwind frees what it had allocated; the worker is checked back in.	**Built and tested** (`tests/tests/cancellation.rs`).
T2 a guest process exits — normally, killed, or killed mid-verb	All three are one path: the guest kernel frees the client root on fd close, so an `RmEvent` arrives, the projection finds no boundary, and the retire is **graph-driven**. Every checked-out worker is cancelled; fresh handles become `Orphans`; the `Proc` lands on the retired list awaiting T3.	**Built and tested.**
T3 the GPU goes idle — THE quiesce point	The guest *tells* us: its driver emits `UNLOADING_GUEST_DRIVER` (RPC fn 47) on an idle release and re-runs the queue handshake. That re-handshake is the quiesce point, and the C already ran its deferred reap there. Same doctrine as the address table: an observed protocol event, never an inference about idleness.	**HALF A LAW.** `reap_retired` exists and `CoreEvent::Deferred(DeferredReap)` exists — **and nothing arms either**. Reap-deferred with no trigger is indistinguishable from reap-never in a run that never quiesces.
T4 the guest driver restarts / the GSP re-handshakes	On `rmmod`, guest panic or VM reset the guest sends *no* `Free` events, so the graph, every live `Proc`, its isolates, arenas, host handles and routing maps all survive into the next driver life — which then derives a second set of components beside the corpses.	**NOT BUILT, and the obvious trigger does not exist.** Bench-measured 2026-07-26: `rmmod` emits **no fn-47 at all** — the idle release at process exit already consumed it. **The unload has no observable signal.**
T5 an isolate worker dies	Routing is settled (the component is condemned, keyed on its client set). Reclamation is not: a worker that died *mid-chain* left allocations that are in no `Orphans` and in no core state. The answer is to escalate to isolate death — `SIGKILL` — because killing the process is how you reclaim state you cannot enumerate.	**Partly built.** The design calls this "the single most leak-prone moment"; the carrier type (`VerbFailure{err, orphans}`) exists.
T6 isolate garbage collection	`SIGKILL` → the process's *exit descriptor* becomes readable → reactor → `waitpid` (non-blocking) → close descriptors, tear down every double-mmap, release arenas, drop worker slots. The kernel already released the host RM objects by closing the connection.	**Designed; the OS half is unexercised** — no real isolate process has ever been spawned.
T7 whole-device teardown	An explicit ordered `shutdown()`, never a `Drop`: stop accepting traps, cancel every worker, wait bounded, retire everything, reap unconditionally, kill every isolate, drain the inbox asserting every remaining event faults rather than mutates, join the threads with no lock held, and **assert the conservation ledger balances**. `Drop` is a tripwire that logs loudly if `shutdown()` was skipped.	**Designed; partly built.**

The instrument all eight are measured by — the conservation ledger `HostLedger`, `kayfabe-mocks/src/lib.rs:1241`

Replays the recorded verb log and demands: every acquisition matched by exactly one release, nothing released that was never acquired — objects *and* mappings.

A test may declare a `ResidueClaim` for what it knowingly leaves outstanding, and the counts are compared for **equality, not as a bound**: less residue than declared fails too, so a fix that closes a leak must delete the claim that excused it.

Its first census over the mean run reported zero leaks — and that was a true negative, not a pass. The workload it was measuring only ever added and removed *RPC* bindings, which own nothing host-side.

Once a phase was written that actually allocates before it frees, the honest baseline was **24 objects, 6 mappings and 24 576 GPA bytes leaking, linear in frees.**

Figure 5: The eight teardown triggers and the retire/reap split. Two of the eight are honestly broken and say so: T3's quiesce trigger is designed but nothing arms it, and T4's obvious trigger was measured on hardware and does not fire.

The lifecycle is where the C artifact's scar tissue is thickest, and it is worth understanding why the split exists. Reaping resources at the moment the guest frees its client root *hangs the dying context's own residual polls* — bench-proven. Not reaping at all exhausts the shared guest-physical window after a handful of process generations — also bench-proven, as bug #80, and *re-introduced and re-found in the Rust core* by the regression test written for it. Hence: retire eagerly (refuse new operations, cheap), reap later at a quiesce point the adapter declares.

Figure 5 marks two of the eight triggers red, and the reasons are different in an instructive way. **T3** is a design that is not wired: `reap_retired` exists, `CoreEvent::Deferred(DeferredReap)` exists, and nothing arms either. **T4** is worse — the trigger the design chose *was measured on hardware and does not fire*. The design assumed `rmmod` emits RPC function 47; the bench showed it emits none, because the idle release at process exit already consumed it. The problem is not that two triggers are indistinguishable; it is that the driver unload has *no* observable signal.

5 The crates

Sizes are `crates/<name>/src/**/*.rs`, measured at `6ce8599`. "Maturity" is this document's reading of the tree, cross-checked against `docs/design/crate_maturity_map.md`.

5.1 `kayfabe-core` — 8 475 lines, **BUILT** and mutation-hardened

Owns the spine. `RmGraph` is the source of truth: a refcounted resource/handle split with `DUP_OBJECT` aliasing, order-tolerant "parked" facts (a mapping that arrives before its target is held, not rejected), mapping refcounts and capacity bounds. `project()` is a pure function from the graph to process boundaries and the two routing tables. `Gpu::apply` is transactional — snapshot, mutate, re-project, roll back on any derivation fault, so a hostile event earns only its own refusal. `gpa.rs` owns the guest-physical window and carves per-process arenas; `GpaSpace::release` takes the arena *by value*, which makes releasing a live process's arena unrepresentable rather than merely wrong.

Deliberately not here: anything about wire formats, any NVIDIA constant, any OS call.

5.2 `kayfabe-abi` — 7 455 lines, **BUILT** — and mislabelled a stub everywhere

Axis A: the per-driver-version wire layouts. Contains an offline generator (`crates/kayfabe-abi/gen`, a separate cargo project with zero third-party dependencies) that parses NVIDIA's own FINN-emitted C headers from the open kernel modules and emits Rust; 11 generated structs plus one hand-transcribed variant; a version-dispatch table keyed on full `major.minor.patch`; and 2 272 lines of tests. The `#[repr(C)]` mirrors are a *layout oracle only* — decode is `from_le_bytes` out of a byte slice, never a cast, so `forbid(unsafe_code)` still holds.

Its test design is the best in the repository: layouts are pinned against **three independent oracles** — NVIDIA's own headers, gVisor's independent transcription, and the C artifact — with a `file:line` citation per number, and a test whose name is `nvos46_the_three_oracles_disagree_and_here_is_exactly_how`. That test found that the C artifact's *parity test* was weaker than the C code it was supposed to guard.

Both of the previous revision's "honest facts" about this crate have expired, and in opposite directions. It now has a consumer — `kayfabe-gsp` depends on it and five of that crate's eight modules use it — so the "7 455 lines with no consumer" finding is closed. And the three places that called it a stub were corrected in `e6ba19e`; `README.md` and `ARCHITECTURE.md` now carry struck-through entries recording the correction, which is the house style working as intended.

What has not moved: `DriverAbiTable` still carries only `version`, `map_dma` and `note`. There is no key for the GSP element layout, so although the tables exist and are version-dispatched, **no production path selects an element layout** — the only callers of `ElementLayout::new` are in `tests/`. The wire tables are a built *capability* that nothing yet asks a question of.

5.3 `kayfabe-linux-raw` — 6 097 lines, **BUILT** — the entire `unsafe` surface

The only crate permitted `unsafe`, and the only one that does not inherit the workspace lints (because `forbid` cannot be relaxed by an inner `allow`). It restates `unsafe_code = "allow"` plus `clippy::undocumented_unsafe_blocks = "deny"`. Every file that uses the keyword is named `*_unsafe.rs`, so `ls` enumerates the audit surface: 7 files, **37 relaxations**, held by a CI ratchet (`EXPECTED_UNSAFE_BLOCKS=37`) that fails on any change in either direction. It also carries a `trybuild` compile-fail matrix — 9 files that must *not* compile — covering things like a bare `MAP_FIXED` address, a page-size literal, and a view outliving its region.

5.4 `kayfabe-gsp` — 3 550 lines, **BUILT** (S0–S5) — **and nothing calls it**

The faked GSP: the guest driver's entire picture of its GPU arrives through this crate. Eight modules. `ram` owns the guest-RAM port and walks the shared region's *self-describing page table* (the region is allocated `NV_MEMORY_NONCONTIGUOUS`, so the C artifact's `base + offset` arithmetic is a latent bug this crate does not reproduce). `ring` is NVIDIA's `msgq` algebra — geometry, cursors, and the `msgqRxLink` acceptance predicate reproduced code-for-code so "would the guest link this header?" is answerable offline. `element` is framing, the 64-bit XOR-fold checksum and multi-element split/join. `boot` is `BootPhase` × `QueueState` with transitions E1–E11, and it deletes all three of the C's one-shot latches — two by construction, one by making the queue GPA exist only inside a `Bound` variant. `rpc` classifies functions into three dispositions (must-reply-synchronously, reply-expected, must-*not*-reply). `replay` is the differential projection its oracle compares, plus a ledger in which a deliberate divergence is admissible only with four things attached, one of them an independent oracle.

Five things it is deliberately not tied to, each with a test that pushes on it: a GPU architecture (every offset is behind `kayfabe_arch::GspModel`; the suite drives the same FSM through two deliberately-different models), a driver version (the element layout is an `ElementLayout` *value*, not a constant), a hypervisor (the only way in is an abstract register access and the `GuestRam` port; CI greps for every VMM type name), a single GPU lifetime (`GspFsm::device_reset` rebuilds the value and teardown drops the queue binding *by value*, so driver-restart works in-process, repeatedly), and a CPU architecture (every wire access is explicitly little-endian).

And it has no caller. §8.2 is about that, and about the two harder problems underneath it.

5.5 `kayfabe-rt` — 2 566 lines, **BUILT** (L1-M1)

The threaded shell. `LockRank{Device=0, Proc=1, Leaf=2}` with an always-on (not `debug_assert`) rank check that panics *before* the OS acquire, so an inversion is deterministic. `SharedDevice` is the plan/execute/commit driver, and it ships **both** lock modes — `Sharded` and `Degenerate` — from day one, tested as first-class configurations with a differential asserting a bit-identical end state, specifically so a late granularity change is never the untested mode.

5.6 `kayfabe-fwd` — 2 491 lines, **BUILT** — and never exercised against a GPU

The entry points an adapter calls: `handle_doorbell`, `publish_backing`, `parse_pushbuffer` (the one parser: CE page-table-write capture, semaphore release, TLB invalidate, everything else opaque and passed through), `forward_engine_object` and `route_control` (the Case-1/Case-2 split), the fence arms, and the present route. This is the crate whose correctness is least established, because its correctness is a claim about NVIDIA's behaviour and nothing here has met NVIDIA.

5.7 `kayfabe-mocks` — 2 160 lines, **BUILT**, test-only

One deterministic fake per port, plus the shared verb recorder and the conservation ledger. Excluded from the mutation denominator on the argument that mutating an instrument is a category error — a surviving mock mutant is not a coverage hole. *Strictly*, it is not confined to dev-dependencies: `tests/` and `fuzz/` list it as a normal dependency, so `cargo build --workspace` compiles it. Nothing shipping links it.

5.8 `kayfabe-vmm-kvm` — 1 891 lines, **BUILT**, **one open defect**

The real KVM memory plane: real `/dev/kvm` file descriptors, real memslots, real vCPU run loops, MMIO exit classification. **Defect #57: the R1 assertion trips roughly 1 in 6 runs under race.** This is why the QEMU adapter has not been started — building a hypervisor adapter on a memory plane with a known lock violation under contention means any later hang has two candidate causes instead of one.

5.9 `kayfabe-isolate` — 1 737 lines, **BUILT** traits; **no real implementation**

The port for the unprivileged host worker: `RmBackend` (12 verbs, all of which must be issuable unprivileged), `Isolate` (two-stage retire, cancellation, quiesce predicate), `IsolateFactory`. `Worker::execute` is the single door to a backend and asserts R1 on entry. A `HostHandle` carries its `IsolateId`, so a cross-isolate handle does not merely fail — it does not type-check.

5.10 `kayfabe-trace` — 1 474 lines, **BUILT** vocabulary; **not threaded**

The typed `TraceEvent` vocabulary, a `TraceSink` port, one-counter total ordering, perf-budget counters and a projection differential. It exists because the GSP milestone's oracle *is* trace replay, so the replay format had to be defined before the harness that consumes it. Two findings from building it are worth carrying: it had to sit *below* `kayfabe-core` (every plane must be able to emit), so it cannot name `RmEvent` or `ProcId` — the bridge is

a trait the owning crate implements with an exhaustive `match`, so a new variant upstream fails the build until the vocabulary names it. And no `&mut Trace` is threaded through any plane signature yet; the vocabulary is driven only from the conformance suite's seam observer.

5.11 `kayfabe-shell` — 995 lines, **BUILT** (L1-M2, in progress)

The reactor: an epoll loop, source resolution, the inbox, and the executor's wake. It carries the restated F1 rule — *assert a bound, never an absence* — as a concrete quantity: after 16 consecutive unproductive wakes it refuses with `ReactorFault::UndrainableSource` rather than spinning. That restatement came from the mutation gate: three spin-versus-park mutants survived precisely because "never busy-poll" is an absence, and an absence is not observable.

5.12 `kayfabe-vmm` — 927 lines, **BUILT** traits only

The hypervisor and display ports: `Vmm` (7 capability groups), `Device`, `Present`. An eighth group — the memory-lock primitive — was *removed* from the trait when it turned out to need no hypervisor cooperation on any backend, which is a good sign about the seam's discipline. **Nothing implements `Device` in production code**; the only two implementors in the tree are test types.

5.13 `kayfabe-completion` — 756 lines, **BUILT**

Per-process queues (pending → in-flight → awaiting-ack → ack), a drain-gated delivery plane whose re-post is driven off *the owner's own poll* (the starvation fix), and mapped-fence arms with the #12 backwards-jump guard. Scored 100% on the mutation gate.

5.14 `kayfabe-util` — 675 lines, **BUILT**

`IntervalMap` (the address table's container, loud on overlap, empty and wrap), a *virtual* `Instant` so no wall clock exists anywhere in the core, the `assert_send_sync!` compile-time gate, and the lock witness that invariant R1 asserts against. The witness lives here rather than in `kayfabe-rt` specifically so that `kayfabe-isolate`, which cannot depend on `rt`, can still assert it.

5.15 `kayfabe-arch` — 757 lines, **BUILT** traits only

The identity vocabulary (`HClient`, `HObject`, `Pdb`, `VChid`, `ClassId`, `GpuVa`, `Gpa`, `GpuId`, `EngineKind`) and the Axis-B port set. It grew by 201 lines at this revision, all of it a new `gsp.rs`: a `GspReg` enumeration and a `GspModel` trait reached through `Arch::gsp()`, which is the seam that keeps every register offset out of `kayfabe-gsp`. The GSP port plan's own §3.5 had recorded "`kayfabe-arch` has no seam for any of this" as a finding; the seam is the finding being paid off. Still zero production implementations: the only `impl Arch` and `impl GspModel` in the tree are `MockArch`, `GspArch` and `FakeGspModel`, all test types. That remains the standing proof that a real one will need no core edits.

5.16 `kayfabe-mmu` — 314 lines, table **BUILT**; **walker is 41 lines of nothing**

`AddressTable` is complete and is the enforcement point for the project's most-cited directive: forward-populate only, unbind eagerly, and **a miss is a fault** — no reverse resolve, no heuristic pick, no MRU fallback exists anywhere in the codebase.

The `walker` module is 41 lines containing one trait and one enum, **with no walk function of any kind**. `WalkResult` is constructed nowhere; `FbRead` is implemented nowhere. The module's own doc comment says the walk algorithm is "also here". It is not.

`tests/` — the conformance suite

43 272 lines of test code against 42 320 lines of implementation. 29 files in `tests/tests/`, 6 more under `crates/*/tests/`, and a `Scenario` DSL in `tests/src/`. The suite is a workspace member with a `[lib]`, which is why `kayfabe-mocks` compiles as a normal dependency. It gained two files at this revision: `tests/tests/gsp_boot.rs` (24 tests) and `tests/src/gspworld.rs` — 1 133 lines of *independent* guest model, written against the driver's own receive path rather than as a script, so that a replay passing because the transition never fired is distinguishable from a replay passing because it did.

6 The invariant catalogue

These are the properties the design is willing to be judged on. Where an invariant has a mechanism, the mechanism is named; where it is only prose, that is stated, because the project's own doctrine is *"prefer a*

mechanism to a sentence — a rule stated only in prose has no reader."

Invariant	Statement	Enforced by
Order-independence	Everything derived is a pure function of declared protocol facts, never of arrival order. "Protocol, not trace."	Permutation/interleave/dup proptests over the whole observable end state (`tests/tests/determinism.rs`).
MISS = FAULT	Tables are forward-populated only; a lookup that misses is a loud fault, never a guess. **Refined:** a ~28-site audit found the absolute had a second, correct answer — the real rule is decided *per site*: *not yet knowable ⇒ DEFER; never knowable ⇒ FAULT*. A mapping arriving before its PDB bind defers; a handle resolving to the wrong *kind* faults. Getting it wrong is asymmetric: fault-when-it-should-defer hangs the guest, defer-when-it-should-fault is a security question.	`AddressTable` has no reverse-resolve path to call. `tests/tests/miss_taxonomy.rs` makes the taxonomy executable. *No shipped site needed a behaviour change; three **doc** claims did, two of them stating the literal opposite of their own code.*
Per-target keying	`Pdb` and `VChid` are per-GPU namespaces. Cross-GPU identical ids are legal; same-target duplicates are loud collisions.	`project()`; `tests/tests/multi_gpu.rs`.
Per-`Proc` isolation (I1)	Identical guest VAs and handles in two processes reach disjoint arenas, disjoint host address spaces and disjoint isolates *by construction*.	Types (`GpaSpace::release` by value; `HostHandle` carries its `IsolateId`) plus a proptest against an injective oracle.
One gated ring path	A doorbell can only reach the host after the #14 working-set gate.	**Now structural, not a call-graph claim.** `VerbPlan::Doorbell` is `#[non_exhaustive]`, so its only constructor is `VerbPlan::gated_doorbell`, which runs the gate; an ungated one is compile error E0639, pinned by a `trybuild` row. The cardinality claim corrected in §10 is what this replaced.
GSP-S1 (new)	**We may never emit an element run wider than the guest's staging buffer.** At driver 580 the guest reads our `elemCount` field directly into an unbounded `portMemCopy` loop whose staging area is exactly `queueElementSizeMax / queueElementSizeMin` elements — 16 on the bench — with the live `msgq` metadata as the very next byte. An `elemCount` of 17 overwrites the metadata the guest is actively using; at the ring's reachable maximum it writes 188 416 bytes past a `portMemAllocNonPaged` allocation. This is a *guest kernel heap* corruption reachable from a value we choose.	**Nothing. It is a design-doc invariant with no code behind it** (`mode2_gsp_port_plan.md` §4.6). The bound holds today only *by derivation* — `MsgLen::new` clamps `rpc.length`, and one encoder computes both numbers — so any future path that sets the field from anything else breaks it silently. The remediation is specified (`gsp_580_correction_brief.md` B2: a named `ElementCountOutOfRange` refusal at the write site, and the same check on receive) and **is not implemented at this revision**: `grep` finds the variant nowhere.
Completion integrity (I2)	A completion fires only from a genuinely-armed source in the *owning* process, at most once; the system forge path is typed so it cannot reach a user process's queue.	Type-level (`Traffic::System`) plus a differential against an independent reference fence model.
Refcount soundness (I3)	A resource is alive $\iff$ it has a live handle or map reference. No premature destroy, no leak, dup survives origin free.	Proptest over deep trees with cross-client dups and interior frees.

Invariant	Statement	Enforced by
DoS containment (I4)	`Gpu::apply` is transactional; every guest-growable table is capacity-bounded with a loud refusal, never OOM.	**Corrected, and the correction is in the design doc:** "no $O(n^2)$ path on hostile floods" is **false and was measured**. The caps bound *memory*, not *time* — see §8.
R1	No blocking host verb may be issued while any ranked lock is held.	`assert_lock_free` at `Worker::execute` and `Worker::with_rm` — the only two doors to a backend. *It does not guard* `Drop`, which is a real call site, and one R1 violation found this week was exactly an `Arc` ownership issue in a `Drop`.
R3	Locks are acquired in strictly increasing rank, at most one per rank.	`check_acquire`, always-on, panics before the OS acquire.
R5	After re-acquiring, re-validate everything the plan assumed.	The four `commit_*` functions; bounded re-plan at `MAX_COMMIT_RETRIES = 8`.
F1	Three unrelated things are called F1 in this project: a fixed threat-model bug, the per-GPU collision guard, and the reactor's bounded-poll rule.	All three real; the overloading is a documentation hazard, not a code one.
GL1–GL13	The guest-memory-lock invariants.	**Documentation only. Zero occurrences in any `.rs` file.** They have no port to be enforced at, because capability group 8 was removed from the `Vmm` trait. Treat as a designed, unimplemented subsystem.

6.1 The two-layer trust model

One normative rule deserves its own space, because it decides designs and because it is the cleanest statement the project has produced:

> **Layer 1 (trap + lock) gives SECURITY against a guest that ignores the protocol. Layer 2 (NVIDIA's own synchronisation points) gives CORRECTNESS for a guest that follows it. Shared tables are NEVER authoritative for an immediate decision.** [docs/design/core_security_threat_model.md §2.1]

Neither layer is a weaker version of the other. Layer 2 alone is not security: a hostile guest can update a shared table *after* validation and *before* we act, and it can fake the handshake entirely. Layer 1 alone is not correctness: a trap says the bytes cannot change while you look; it does not say they are *finished*. A cooperating guest halfway through publishing a page-directory update has written something internally inconsistent and perfectly stable, and reading it under a perfect lock yields a perfectly-locked wrong answer.

The consequence — *a shared table is evidence, never authority* — is why the address table is forward-populated and why a miss is a fault.

And the C's measurement sharpened it in a way worth carrying. The Layer-2 edge is *whichever* edge the protocol commits at, not a favourite one. On the kernel/UVM/RM paths that edge is the TLB invalidate. On the GSP-emulated compute path it is *not*: round 6 of the #14 investigation measured **zero** occurrences of both invalidate transports there. A design that had encoded "read at the invalidate" as *the* Layer-2 rule would have had no synchronisation point at all on the path that matters most. The general rule survived; one of its instances did not.

What the C artifact does not have is Layer 1. There is no lock of any kind between QEMU and an isolate in a system that runs 22 real GPU applications at host parity. It ships a copy-once snapshot instead, and the reasoning is written out in its source. The working system to date has been running on Layer 2 alone — which is correct for a cooperating guest and is not a security property.

And the boundary points both ways, which building the GSP made concrete. Everything above is about not trusting the guest. **GSP-S1** is the mirror: the guest kernel is inside the boundary we are defending, and the faked GSP is a device that can corrupt guest kernel memory from a width *we* choose. The mechanism is fully traced in `mode2_gsp_port_plan.md` §4.6 and reduces to one sentence — at driver 580 the guest reads our `elemCount` field into a copy loop bounded by nothing but the ring, and the byte after its staging area is the `msgq` metadata it is actively using, so the corruption is immediately self-amplifying. It is the same defect class as the C artifact's unguarded `% s->q_msgcount SIGFPE`, pointed at the guest instead of at QEMU. The obvious objection — *"the guest is the victim, so it is the guest's bug"* — is answered in the plan and the answer is that it is both. **At this revision the invariant is written down and not enforced**; that is the most security-relevant thing in this document that is a sentence rather than a mechanism, and by the project's own doctrine a rule stated only in prose has no reader.

7 What is tested, how, and what that does and does not prove

7.1 The shape of the suite

Measurement	Value	How
Tests reported green	**547**	`KAYFABE_NO_KVM=1 KAYFABE_SLOW=1 cargo test --workspace`, recorded in commit `e508a8a` — [unverified] here, since no `cargo` was run for this revision either
`#[test]` attribute sites	544	grep over workspace members; excludes the detached generator and the `trybuild` ui fixtures
`proptest!` blocks	16	each is one test that runs many cases
Checked-in proptest failures	**0**	`find` for `proptest-regressions`: nothing exists
KVM-gated tests	36	CI floor asserts ≥ 35 were *reached*
TSan targets	4 files, 68 tests	nightly only
Fuzz targets	1	`parse_pushbuffer`; CI compiles it, does not run it
Implementation lines	42 320	`crates/*/src/**/*.rs`
Test lines	43 272	`tests/`, `crates/*/tests/`, `fuzz/`

The suite runs in about 20 seconds with no GPU, no OS dependency and no wall clock. That is a deliberate and load-bearing property, not a convenience: it is what makes the adversarial suites affordable enough to run on every change.

7.2 The six CI gates

There is one workflow file with six jobs, and the `stable` job grew from eleven steps to twelve at this revision. The README's "four standing CI gates", flagged as stale by the previous revision of this paper, has since been removed from the README (§10 row 5).

Job	When	What it asserts
`stable`	every push + PR	Twelve steps: build; **test with KVM forced absent**; assert the KVM-gated tests were *reached*; clippy `-D warnings`; `fmt --check`; the hexagonal-boundary grep; the VMM-vocabulary grep; **the generation-name grep** (no concrete chip or driver version in the ten logic crates — added in `a0a5834`, and it is the gate that makes `kayfabe-gsp`'s version-agnosticism mechanical rather than aspirational); the `unsafe`-surface `ls` gate; the GPA-accessor gate (exactly one call site — and `kayfabe-gsp` is on its list, which is why that crate has a `GuestRam` port of its own instead of touching `Vmm`); three unsafe-containment asserts including the ratchet at 37; and `fmt --check` for the fuzz workspace.
`aarch64`	every push + PR	`cargo check --workspace --target aarch64-unknown-linux-gnu`
`nightly-fuzz`	every push + PR	the fuzz harness *compiles*
`slow`	nightly	the full suite with `KAYFABE_SLOW=1`
`tsan`	nightly	ThreadSanitizer over four threaded targets, `-Zbuild-std` so std's own synchronisation is instrumented
`mutants`	weekly	`cargo mutants` over every production crate

7.3 Exercised versus merely present — the project's own core doctrine

This distinction is the most useful thing in the repository, and it is worth reproducing the incident that generated it, because it is the template for every criticism in §8.

> The conservation ledger's first census over the mean run reported **0 leaked objects** — and that was a **true negative, not a pass**. A design document claimed the workload "exercises T0 thousands of times per run"; the churn it actually drives adds and removes *RPC* bindings, which own nothing host-side, and the script's two real subset-frees targeted channels that phase 0 deliberately leaves virgin. **The path had never been reached.** Reaching it took a new test phase. Once reached, the honest baseline was **24 objects, 6 mappings and 24 576 GPA bytes leaking, linear in frees.** [docs/design/testing_doctrine.md §1]

Three rules follow, and the project applies them: a reclamation test that does not first allocate proves nothing; every instrument needs a non-vacuity assertion (a case where it is known to read non-zero); and **bite-check the instrument, not only the fix** — reverting the fix must move the number, and if it does not, the number is not measuring the fix.

Bite-checking is real practice rather than aspiration, and the evidence is that the repository records bite-checks that *did not bite*: at least six such notes are written into the source at the site they concern, each explaining what the neutering failed to move and what was added as a result. One has been promoted into a permanent test that plants a gap and a repeat in a trace and requires the order checker to fire. A commit message from this week reads, in part, *"the bite harness itself lied twice."*

The doctrine also produces two mechanisms directly. Slow tests are gated by `KAYFABE_SLOW`, and when unset they print `SKIPPED (slow): …` **straight to stderr**, bypassing libtest's capture; nothing in the suite is `#[ignore]`d, on the argument that `#[ignore]` is invisible in a summary and impossible to count. And the KVM gate prints on *both* arms — `KVM-GATE: RAN` or `KVM-GATE: SKIPPED` — with CI asserting a floor on the sum, because the reverse failure (all 36 gated tests silently vanishing while CI stays green) is equally silent.

7.4 The mean test, and composed versus isolated runs

`tests/tests/l1_mean.rs` is the largest single test file (35 tests) and is designated *the arbiter*. Its standing is: *pass = the design survived contact; fail = the doc changes, not the assert.* The rule it exists to enforce is an owner directive with four obligations — drive realistic multi-process × multi-thread × multi-GPU traffic end to end through the mocks and assert the observable end state; make it mean rather than happy-path (free mid-flight, race a re-declaration against a still-publishing generation, kill a worker partway, run both lock modes); wire every new milestone into it; and **never narrow a test to make it pass**.

The justification is measured, not asserted. One identity round landed 7 bite-checks; **none of the 363 pre-existing tests caught any of them**. Every real defect that day came from a composed run or a newly sharpened criterion. In the doctrine's words: *"the happy path found nothing, all day."*

A representative finding: a test named `worker_death_retires_the_proc_loudly_and_never_resurrects` was green — and green *only because it never issued an apply after the HUP*. The composed run put a worker hangup and an alloc/map-heavy workload in the same run and found that an out-of-band retire is undone by the next refresh: a guest able to crash its isolate worker got a **clean new isolate** on its next RM event. The first fix for the related leak class was a use-after-free of its own making.

7.5 The conservation ledger

The instrument all eight teardown paths are measured by. `HostLedger` replays the recorded verb log and demands every acquisition matched by exactly one release and nothing released that was never acquired — objects *and* mappings. A test may declare a `ResidueClaim` for what it knowingly leaves outstanding, and the counts are compared **for equality, not as a bound**: *less* residue than declared also fails, so a fix that closes a leak must delete the claim that excused it. There is deliberately no `allow_leaks` escape hatch.

Its stated limit, recorded in the source: the ledger increments only on *success*, so it says nothing about failed verbs. That is why a separate `VerbFailure{err, orphans}` type exists to carry handles out of a chain that died partway.

7.6 The mutation gate — and why no score should be quoted today

The gate mutates the code and measures whether the suite notices. It has twice been the most valuable instrument in the project and twice been silently wrong *upward*, which is the failure mode the project cares most about.

The first lie: it tested nothing. `kayfabe-fwd` and `kayfabe-rt` have no unit tests of their own; without `--test-workspace true`, cargo-mutants ran an empty suite and reported everything missed.

The second lie: compiler crashes vanished from the denominator. cargo-mutants rewrites one file per mutant in a copied tree that reuses one incremental dep-graph cache, and that cache does not survive the churn — rustc ICEs. cargo-mutants cannot tell a compiler crash from a type error, so it files the mutant *unviable* and **silently removes it from the denominator**. Measured: **136 of 293 mutant builds ICE'd**, turning a real 88.95% into a reported 67%. CI now fails on any ICE in a mutant log, and on a zero viable count, *before* the threshold is read.

The historical numbers, each true of the scope it was measured on: **99.2%** (245/247) on the pure L0 core, with both residuals argued equivalent or acceptable; **92.44%** (159/172) on the L1 threaded surface, where the residual untested *logic* is 2 mutants wide and named.

Do not quote a current mutation score. On 2026-07-27 the scope changed from four hand-picked paths to every production crate — the old scope covered roughly 41% of the implementation, and *every real defect that week lived in the unmeasured part*. The CI threshold of 91% was derived against the old scope, so it now describes a different population. The workflow marks it **PENDING RE-DERIVATION** in its own text. A full-scope run has been measured once at **90.60%** raw, rising to roughly **94.9%** once test scaffolding is excluded from the denominator (48% of that run's misses were mutations of mocks), but neither number has been confirmed by a second run.

And a second reason not to quote one at this revision. The scope is `-f 'crates/*/src/**/*.rs'`, so it now covers `kayfabe-gsp` — 3 550 lines that did not exist when either number was measured, and roughly 8% of the implementation. *No mutation campaign has ever run over them* [unverified]. Conversely, the GSP port plan's own §5 warns that "the mutation job's `-f` scope does not include the crate"; that warning was written against the pre-`0b78102` four-path scope and **is itself stale** — the scope was widened before the crate was written.

7.7 The OS-free configuration, and why it exists

`KAYFABE_NO_KVM=1` forces the KVM probe to report the device absent, so the pure logic core plus mocks — no OS layer at all — runs on purpose on any box. CI runs *this* as its default test configuration.

The reason it was made explicit is the doctrine applied to itself. GitHub runners have no `/dev/kvm`, so CI was already running the OS-free configuration — **by luck of what the runner image happens to lack**. It was unreproducible on any developer box, and it would have lapsed silently the day a runner image started shipping the device. In the commit's own words: *"a guarantee that holds by luck is the same shape as a green instrument on an unexercised path."*

The owner's question that prompted it is worth repeating because it is the right question: *"I hope you didn't delete the mocks so the core can still be tested without OS layer? and that must pass as well so we don't lock in architectures/os semantics/qemu and later regret."*

7.8 What the suite does not prove

- **It does not prove anything about NVIDIA.** Every test runs against `MockArch`, whose encodings are deliberately *not* NVIDIA's. That is the right design for proving the seam, and it means the suite cannot catch a wrong belief about the real protocol. Only the C artifact and the bench can do that.
- **The with-KVM configuration has no automated gate.** CI runs *only* the KVM-absent configuration. The 36 tests that use a real `/dev/kvm` — including the entire real memory plane and the vCPU/MMIO dispatch — run on developer boxes and the bench, and nowhere else. This is the largest hole in the CI story, and the workflow says so in its own text rather than hiding it.
- **The fuzzer is compiled, not run, by CI.** One target, over the one place raw guest bytes reach the core. Its two historical findings were minimised into deterministic in-tree regressions, which is the stronger form. The corpus is git-ignored, so a fresh clone starts from empty.
- **Single-threaded logic testing of a concurrent system.** The core is tested deterministically because it is a pure state machine; the concurrency is tested separately by the threaded suites plus TSan. The argument is sound, but it depends on the core genuinely having no interior mutability — which is asserted at compile time, and is the sort of property worth re-checking if you are looking for holes.
- **The page-size axis is designed and not built.** The testing doctrine cites `[4 KiB, 16 KiB, 64 KiB]` as an application of its "both modes ship from day one" rule. The environment variable that would select it does not exist in any `.rs` file — only in forward-looking comments. The `LockMode` half of that same rule *is* built and exercised.

8 Discussion: the good, the bad, and the risky

8.1 What is genuinely good

The purity constraint is real and it is mechanised. A "pure core" claim is usually aspirational. Here it is three CI greps, a manifest-level `forbid`, a naming rule that makes `ls` the audit tool, and a ratchet on the exact number of `unsafe` relaxations. The external dependency surface is three crates, two of them dev-only.

The failure ledger is unusually honest and it is load-bearing. Two design documents carry "contact logs" — `l1_concurrency.md` §12 and `l1_os_shell.md` §14 — recording every place building the code proved the design wrong. They are long. They contain entries whose titles are things like "the criterion was wrong", "boundary-1 broken", "two sections of §3 contradict each other", "the design was wrong about its own severest claim". A project that writes these down is a project whose other claims are worth more.

The adversarial suites found real bugs before hardware existed. 15 at L0, more since at L1, including two control-plane *wedge* bugs reachable by a hostile guest process — a parked `SET_PAGE_DIRECTORY` and a parked `MAP_MEMORY_DMA`, each of which could be aimed via a dup alias to make a benign third party's allocations fail forever. Both were found by a property test, not by inspection.

Both lock modes and both fallbacks ship as first-class tested modes. The argument is sharp: the slow path is the path the system takes *when something has already gone wrong*, and a fallback exercised only by the fast path declining has test coverage proportional to how often the fast path fails — which is the quantity the optimisation exists to drive to zero.

The ABI oracle design. Pinning generated layouts against three independent readings, with a test that documents exactly how they disagree, is better than what most projects do, and it caught the C artifact's own parity test being weaker than the code it guarded.

8.2 `kayfabe-gsp`: boot is modelled, reboot is not, and one open item has no proposed resolution

> **This section replaces one that said "34 lines, and it is the critical path".** That sentence was true at the previous pin and is not true now: `f2055bf` built stages S0–S5, 3 550 lines across 8 modules, with 24 dedicated tests, an independent 1 133-line guest model, and a composed run inside the arbiter suite. The risk did not shrink so much as *change shape*, and the new shape is more interesting than the old one. It is also not a victory lap: three of the plan's nine stages are unbuilt, one open item has no proposed resolution at all, and the plan the crate was built from was written against the wrong driver version.

What is built, and what "built" is worth here

The five seams the crate claims — GPU architecture, driver version, hypervisor, GPU lifetime, CPU architecture — each have a test that pushes on them, which is the difference between a seam and a decoration. Two are worth singling out because they are cheap to fake and were not faked. The lifetime seam is exercised by running *repeated* driver lives in one process: the C artifact's measured second-life failure (`msgqRxLink` timeout, `-7`, 71 064 retries) has exactly one cause in the driver's own source, `rx.size != size`, and the FSM is shaped so that cause cannot arise. The architecture seam is exercised by driving the same FSM through *two* deliberately-different register models, which is the same argument as `MockArch` one layer down.

The oracle is the part to be sceptical of, and the crate says so itself. The design called for trace replay against the C emulator's recorded boots. **Those traces do not exist**: the capture patch is ∼60 lines against a C file this work does not own, and it has not landed. So S5 is proven against a *guest model* — an independent implementation of the driver's receive path — which is a stronger oracle than a synthetic script for everything except the one thing it was supposed to establish. Until the capture lands, **"byte-equivalent with the C" is untested**.

Boot is modelled. Reboot and resume are not.

The plan has nine stages. S0–S5 — the six that need no GPU — are the boot and steady-state protocol, and they are what `f2055bf` built. **S6, S7 and S8 are the three that need a GPU, and none is built**; S6 is additionally blocked, because `4a0cb29` records the bench host as offline at the provider. Two of the plan's open questions (**O3**, whether the sequence numbers reset across a true `rmmod`/`insmod`; **O5**, which boot mode a post-teardown re-acquire uses) are answerable *only* by a bench run, and O3 is not even reachable before S7. This matters

more than the stage count suggests, because the owner's standing requirement is that a GPU restart — idle, released, re-acquired — must work *with no bolt-on*, and restart is exactly the region S6–S8 covers. Boot is the part that could be modelled offline, so boot is the part that got modelled.

The sharpest open item: at 580 the resume handoff is an RPC *we* would have to send

This is §11-O7a of the port plan, and it is the one item in the whole component with **no proposed resolution**, because the evidence that would settle it is not in any open-source tree.

The mechanism is present at both driver versions and reads identically: a core resume resets into RISC-V, reprograms the LibOS boot-args address, starts SEC2 and waits on `NV_PGC6_BSI_SECURE_SCRATCH_14._BOOT_STAGE_3_HANDOFF`. What differs is **who initiates it**:

	580.159.04 (the bench)	**610.43.02 (the vendored spec)**
where the wait lives	inside the CPU sequencer executor, as opcode `CORE_RESUME`	promoted to a first-class HAL, `kgspExecuteCoreResume_TU102`
how it is reached	**only** through `_kgspRpcRunCpuSequencer` — i.e. only if *we send* a `GSP_RUN_CPU_SEQUENCER` event	locally, from two call sites in the driver. **Nothing is required of us.**

Three things follow, and none of them is comfortable.

It is not a layout difference, so a version table does not absorb it. Every other 580-vs-610 divergence the project has found is an offset, a field count or a constant — one row in a version-keyed table. This one is a difference in *who initiates*, which means a faked GSP that must emit sequencer buffers at 580 and must not at 610 has version-conditional *behaviour*. `docs/design/four_axes_of_variation.md` §5 rule 1 forbids exactly that: *"No* `if guest_version == …` *in a logic crate. Transitions fire on observed protocol facts, never on driver identity."* The two honest options named in the correction brief are a version-keyed *capability* on the FSM or a loud refusal, and the brief declines to choose between them on the grounds that choosing requires an answer it does not have.

The answer cannot be read out of the open source. What the open tree shows is that the path exists and how it is triggered. What it cannot show is whether the firmware ever triggers it on a path we must support, because the *contents* of a sequencer buffer come from GSP-RM firmware, which is not open. Specifying an emitter now would be guessing, and the plan says so rather than shipping a guess.

So the requirement is genuinely at risk, and this is the cleanest statement of it in the project. If any restart scenario in scope reaches `CORE_RESUME` at 580, then a faked GSP that never emits `GSP_RUN_CPU_SEQUENCER` cannot drive it, and "GPU restart must work with no bolt-on" is not currently satisfiable by the design as written. *Whether* such a scenario exists is unknown [unverified]. The next step is a reading task, not a hardware task — enumerate the callers of consequence of the sequencer executor at 580 — and it is recorded as O7a rather than resolved.

One further sharpening worth carrying, because it cuts against the project's own instinct: the relayed finding that started this said "580 has a resume handoff 610 never reads". **That half was wrong** — 610 has the identical routine reading the identical register. The correction log reports it as wrong rather than quietly fixing it, which is the behaviour a reviewer should be checking for.

The plan was written against the wrong driver, and one item was RESOLVED backwards

This is the most instructive thing that happened to this component, and it is not flattering.

The port plan took its NVIDIA citations from the vendored open kernel modules at **610.43.02**. **The bench runs 580.159.04.** The two do not merely differ in offsets; they differ behaviourally, and 3 550 lines were built against the un-caveated plan before anyone vendored the matching tree. Eight claims changed when `ogkm-580.159.04` was finally vendored (`7af23cb`): the element count is *read from a field* at 580 and *derived from a length* at 610; MCTP/NVDM transport headers do not exist at all on the 580 GSP path; the layout break interval narrowed from `(570, 610]` to `(595.84, 610.43.02]`; the init RPCs are queued *before* bootstrap at 580 rather than inside it; the suspend sentinel is an exact equality at 580 and a mask at 610; and the queue geometry is compile-time at 580 rather than negotiated, which removes the "the guest declares its own geometry" argument the plan had presented as its highest-leverage design choice.

The worst instance is §11-O7, which had been marked RESOLVED with the answer that is backwards for our bench. It said `GSP_RUN_CPU_SEQUENCER` "is not implemented, do not emit it" — true at 610, where the executor was deleted; at 580 it is fully implemented, dispatched, and first on the bootup-window allowlist,

so an emitted sequencer buffer would execute register writes and falcon resets inside the guest kernel rather than being harmlessly ignored. The RESOLVED status has been withdrawn.

The class, which is the part worth keeping. A *versioned* specification was cited as if it were *the* specification. The fix is mechanical and now normative: every citation carries its tag (`ogkm-580: / ogkm-610:`), an untagged claim is unverified by definition, and line numbers *and paths* never cross tags — the whole `msgq` library moved directories between the two. **That fix is not applied to the code yet**: the workspace's crates carry **121** bare `ogkm:` citations in doc comments, 96 of them in `kayfabe-gsp` alone, and every one is a 610 path with a 610 line number — several naming files that do not exist at 580, since the whole `msgq` library moved directories. The CI grep that would prevent the recurrence is specified and not written.

The self-critical note in the correction log is the one to read: one claim in the same document was marked `[inferred]` and turned out to be right, while O7 was marked RESOLVED and turned out to be backwards — and the difference was not luck. *The inferred claim knew it was second-hand and the resolved one did not.*

What this does and does not change about the critical path

The gap moved up one layer; it did not close. There is still **no production caller**: nothing outside `tests/` implements `GuestRam` or `Device`, and the crate's output type `RpcCommand` has zero references anywhere else in the tree, so the `RpcCommand` → `RmEvent` bridge — which the plan deliberately places in `kayfabe-fwd` rather than guessing it here — does not exist. Nor does any production selector for the wire layout: `DriverAbiTable` has no GSP element key, so the version-dispatch machinery that makes the crate version-agnostic has, at this revision, nothing production-side asking it a question.

The critical path is now the L2 QEMU adapter (§8.7), which is the thing that would give `kayfabe-gsp` a caller, guest RAM to read, and a BAR to answer from — and which has not been started, deliberately, because `kayfabe-vmm-kvm` still carries defect #57.

8.3 Nothing has ever touched a real GPU

Stated once more because it is the thing most likely to be forgotten between here and the end of the paper. There is no binary. `Device` has no production implementor. `RmBackend` and `IsolateFactory` have exactly one implementation each and it is the mock. No isolate process has ever been spawned. The composition of threads that would constitute a running system exists only inside one test file.

The faked GSP does not change this and it is worth being explicit, because it is the component most likely to be misread as hardware contact. `kayfabe-gsp` parses real `msgq` headers, real element framing and real RPC envelopes — but every byte it has ever parsed was written by a test into a `BTreeMap` behind the `GuestRam` port. No guest driver has ever posted to it. Its own oracle, trace replay against the C emulator's recorded boots, is **not available**: the capture patch has not landed, so the traces do not exist, and the crate is proven against an independently-written model of the guest rather than against the guest. Two of its remaining open questions (**O3**, **O5**) are answerable only by a bench run, and the bench host is currently offline at the provider.

What this specifically means for the claims in this document: everything about *structure* (the DAG, the purity, the type-level isolation, the lock discipline) is verifiable today and this paper verified it. Everything about *behaviour against NVIDIA* is inherited from the C artifact and re-expressed, and the re-expression has not been checked against the thing it models. The gap between "our mock says the core does the right thing" and "a real driver agrees" is exactly the gap the C artifact spent two months and 739 commits closing, once, in C.

8.4 The quadratic control plane — an open, measured DoS

The invariant catalogue's I4 says hostile input is *contained*: no wedge, no OOM, no bystander corruption, every hostile event earning only its own refusal. All of that is property-proven. It says nothing about *cost*, and the project measured the cost and recorded the result against itself:

> The control plane is O(live objects) per event, so N events cost $O(N^2)$. Measured on a debug build: **1 000 events 0.85 s; 8 000 events 54.8 s**. The capacity cap is 2^{18} live handles, so a guest can drive that curve deliberately — and **PyTorch startup reaches it benignly**. Deleting the per-event graph clone saves ~24% and leaves the curve quadratic, so the clone is neither the cause nor the fix.

The stated statement in `ARCHITECTURE.md` — "no $O(n^2)$ path on hostile floods" — is marked **false** in the same document. Two candidate fixes are named and neither is small: incremental derivation, which is a redesign of the "derived state is a pure function of the graph, never accreted" decision that the entire determinism

differential rests on; or an undo journal in the most safety-critical file in the repository. A sibling of the same species *was* fixed (a condemned-list carry-forward, $O(n^2 \log n)$, 55 s $\rightarrow$ 3.8 s, by union-find), which is evidence the class is tractable and also evidence that it recurs.

8.5 Reclamation is the least finished plane

Figure 5 marks two triggers red. Restating the sharper form:

"Retire eager, reap deferred" is currently half a law. The deferral has no bound and nothing arms it. In a run that never quiesces, reap-deferred and reap-never are indistinguishable.

The one genuine cross-process reference has a disposition that no ledger will like. On a kernel/UVM dup of a user address space, refcount zero and per-object `Free` are not the same event: the last reference retires and reaps the owner *without issuing a single* `Free` *verb*, and the disposition of record is the isolate process's death. GPA is conserved independently and asserted. But any future reclamation ledger, quota or accounting hook will be wrong on that path first — which the design says, in advance, in its own words.

And the most leak-prone moment has no vocabulary at all until L1-M2 lands. A worker that dies *mid-chain* cannot run its own unwind, so everything allocated before the failure is in no `Orphans` and in no core state. It is unrecoverable and no current test can see it. The design's answer — escalate to killing the isolate, because killing the process is how you reclaim state you cannot enumerate — is correct and is not yet built.

8.6 arm64 is measured for the core and unmeasured for the shell

The full suite *ran* on a GB10 Grace-Blackwell aarch64 host: **372 passed, 0 failed**, identical to x86_64. That is a real upgrade over a cross-compile check, and it covers the concurrency suites, whose memory-ordering behaviour is exactly what a `cargo check` cannot see.

What it does not establish, stated by the project so nobody over-reads it: `getconf PAGESIZE` was 4096 on that host, so the 16/64 KiB page-size rule — the one real pressure point — is still unexercised. The box was a container with no `/dev/kvm` and no `/dev/userfaultfd`, so **no L1/L2 OS-shell behaviour was exercised on ARM at all**. And there are zero VM-capable arm64 machines on the bench provider, so settling the remaining question by experiment requires hardware from somewhere else.

The honest summary is the project's own: *the pure core is arm64-clean by measurement; the OS shell on arm64 is entirely unmeasured*.

8.7 The QEMU $\geq$ 10.2 floor buys less than the decision that took it assumed

The design requires `memory_region_enable_lockless_io()`, which lands in QEMU 10.2. That buys a genuinely important thing — MMIO dispatch without the big QEMU lock, which is what makes the doorbell path viable — and upstream's helper also sets the reentrancy guard in the same body, so a discipline the design planned to maintain is no longer ours.

The floor decision's second claim was economic: requiring a version rather than patching one deletes *"the whole 'which QEMU are we built into' ambiguity"*. **The L2-Q adapter design** (`b31487a`) **checked that claim against QEMU's source and it does not hold.**

> **QEMU v10.2.0 has no supported out-of-tree device mechanism, so our device must be compiled inside a QEMU source tree.** Two mechanisms, both closed: `module_load_qom()` resolves a QOM type name against `module_info`, a table *generated at QEMU build time*, so a `.so` that is not in that table can never be discovered by type name — and `QEMU_MODULE_DIR` changes only the directory searched, not the table. `module_load_dso()` additionally requires the module to export a per-build `CONFIG_STAMP` symbol; upstream's own failure hint is verbatim *"Only modules from the same build can be loaded."* [v10.2.0 util/module.c, include/qemu/module.h]

State it exactly, without overstating it. The floor *does* delete a patch we would otherwise have had to carry: the cancelled 9.2 backport modified upstream semantics inside `system/physmem.c`, had to be re-derived every release, and a mis-rebase would silently change how *every* device in the machine dispatched. What replaces it is an additive overlay — a directory dropped into a stock 10.2.x checkout plus two hunks (one `meson.build` line, one `Kconfig` stanza) — whose only conflict surface is a build failure, never a behaviour change. That is a real improvement. **What it does not buy is "install your distro's QEMU and run".** The user builds QEMU once. Three of the four costs the floor decision itemised against the backport — a rebase every release, a build script that must reproduce it, and a supply-chain claim we make to every user — still apply.

There is exactly one path to a genuinely stock QEMU, it is named, and it is not taken for v1. `vfio-user` would make our device a separate process speaking a documented socket protocol to an unmodified distro-packaged QEMU. It is present at v10.2.0 (`hw/vfio-user/`, two documentation files, a vendored `libvfio-user`) and

`MAINTAINERS` lists it `S: Supported`. It would delete the QEMU tree, the overlay, and the second `unsafe` crate the in-tree route needs. It is rejected for v1 on one source read: nothing in `hw/vfio-user/` calls `memory_region_enable_lockless_io`, so a trapped BAR access becomes a socket round trip *taken with the BQL held* — reintroducing and amplifying the exact serialisation the floor was taken to remove. The design records the small experiment that would change the verdict (measure the round trip both ways, and ask whether marking the client's regions lockless is a patch upstream would accept) rather than closing the question.

The facility inventory that priced the floor also **reversed two of its own design assumptions**, which is worth knowing before trusting the rest of it: the read-only-memslot region-lock fallback, priced in the design at 1.49 µs as "a flags-only flip", is *unreachable through QEMU* — QEMU's flatten pass turns a readonly change into a delete-plus-add. And checkpoint-restart (`cpr-transfer`) is a lifecycle event the design had never considered.

One more cost the adapter design surfaces and does not argue away. A QEMU adapter needs `extern "C"` entry points, raw `MemoryRegion *` handles and adoption of a host pointer QEMU owns — all `unsafe`. The workspace's CI gate exists to keep the audit surface at *one* crate readable in one sitting, and its own failure text says adding a second is "a design decision, not a manifest edit". L2-Q takes that decision: `kayfabe-qemu-raw` becomes the second audited `unsafe` crate, and the ratchet moves from a constant to a named two-element list. The acceptance criterion the gate protects becomes "read two crates". That is the honest price and the design says so in those words.

8.8 The bench is a single serialised GPU

All hardware work happens on one rented GPU box, strictly serially: concurrent or mid-ioctl activity wedges the GPU. The C artifact needs a **fresh VM boot per clean run** — and, per §3, that discipline turns out to *be* a bug rather than a precaution. Two cargo builds on one box corrupted a mutation run through the shared `~/.cargo`, so the rule is one cargo per box.

The consequence for the reader: every hardware number in this document is $n = 1$ in the statistical sense and comes from one GPU model (GA106/GA107 class consumer Ampere) on one driver family. There is no second machine, no second GPU generation, and no continuous hardware testing.

8.9 The recurring failure mode: claims that were true *once*

This is the pattern the project keeps hitting, it has a name in the repository, and it is the single most useful thing for a reviewer to internalise.

> Across three independent passes, roughly **twenty** instances were found of documentation asserting more than the code did — **several stating the literal opposite of their own code**. **Every one of them was true when written.** That is the point: this is not carelessness, it is entropy. [docs/design/testing_doctrine.md §6]

The instances are specific and they are not small:

- A design document named `memory_region_clear_global_locking()` as the mechanism for a decision. **That QEMU function had been deleted five years earlier.** The reference document written to price the floor now carries a `file:line` on every row for exactly this reason — and found two *more* live instances of the same decay while being written.
- A refactor plan cited a userfaultfd proof-of-concept file **that does not exist**. The mechanism was planned four times and never built; its demand-fault path is dead code that calls `stub_exit(139)` and never returns, so the fault address has been permanently zero since a commit early in the project's life. Nobody noticed because "planned" and "presumably done" read the same in a plan.
- **Two sections of one design document contradicted each other**, found only when the code between them was written.
- Two claims about the C's behaviour, cited from the C's own source comments, turned out to be **false when measured on the bench**. The rule extracted: citing a comment is citing a belief.
- A rustdoc described a default-to-GPU-0 guess *inside the one resolver whose entire discipline is "never guess"*. Another claimed "MISS=FAULT is preserved — never a silent skip" while its own body correctly skipped on both arms; the body was right and the sentence was wrong about the most-cited exception in the codebase.

The project's response is the right one: prefer a mechanism to a sentence. A rule stated only in prose has no reader. `HostHandle` carrying its `IsolateId` is the same rule with a compiler behind it. The house style for

corrections is *strike visibly* — keep the original, struck, with the correction and its evidence — so a reader who remembers the old claim learns it was wrong rather than doubting their memory.

And the pattern is live right now, in this document's subject, and in this document. The first revision of this paper found fourteen instances in the Rust repository (§10 rows 1–14). Four days and fourteen commits later, this revision found six more — **two of them in the first revision's own text and figures**, including a dependency matrix that omitted one of the edges it claimed to derive from the manifests. Every single one of them, in both passes, is a claim with *no gate behind it*. Every claim that *does* have a gate — the boundary greps, the generation-name grep, the ratchet at 37, the reached-count floor at 35, the "exactly one GPA call site" assertion, and now the E0639 compile-fail row that makes the ring gate structural — is mechanically true today. That is either a devastating footnote or the strongest possible evidence for the project's own thesis, and it is the latter.

The strongest single instance is in §8.2 and it is worth restating here as a pattern rather than a fact about NVIDIA. A specification was cited as if it were the specification, when it was one version of a specification; 3 550 lines were built against it; and one question had been closed with an answer that is exactly backwards for the machine the code will first run on. The claim that survived that audit was the one marked `[inferred]` — because it knew it was second-hand. The claim that did not survive was marked **RESOLVED**.

8.10 Risks this document adds

- **Test code slightly exceeds implementation code, and the mocks are the only implementation of three ports.** The suite is therefore testing the core against a model of NVIDIA written by the same author as the core. `MockArch`'s deliberately non-NVIDIA encodings mitigate the *encoding* half of that; they do not mitigate the *semantics* half. If a belief about RM semantics is wrong, the suite will confirm it consistently.
- **The project is four days old and moves faster than its own documentation.** 107 commits in 96 hours, with several design documents rewritten mid-build. That is what produced the stale claims in §10, in a repository that has an explicit doctrine against stale claims. It has already produced more: fourteen commits after the first revision of this paper, six further instances existed, two of them in this paper. The rate is not slowing, and no gate covers prose.
- `kayfabe-vmm-kvm` **defect #57 is a lock-invariant violation under race, and it is blocking by design.** That is the correct call. It is also the kind of defect that is cheap to live with and expensive to leave: an R1 violation that fires 1-in-6 under contention is exactly the shape that becomes a rare production hang.
- **The security model has been red-teamed at the logic layer only.** Memory safety, out-of-bounds and host-breakout are explicitly deferred to a post-L2 audit, on the sound argument that they are born at the mmap/isolate/trap surfaces and there is nothing to audit in pure logic. Those surfaces now partly exist (`kayfabe-linux-raw`, `kayfabe-vmm-kvm`), so the deferral is beginning to age.
- **The** `Vmm` **port's hypervisor-agnosticism is contracted and CI-gated but has exactly one adapter.** "A second backend costs one crate and zero trait changes" is a claim that only a second backend can settle.
- **The one guest-facing memory-safety invariant the GSP work produced (GSP-S1) is prose, not code.** An `elemCount` we emit above the guest's staging bound corrupts guest kernel heap. The bound holds today only because one encoder happens to compute both the length and the count; nothing checks it at the field. The remediation is specified and unimplemented. This is the project's own "prefer a mechanism to a sentence" rule, unpaid, on the highest-stakes sentence it has.
- **One open item bears directly on a stated product requirement and has no proposed resolution.** §11-O7a (§8.2) may mean that "GPU restart works with no bolt-on" is not satisfiable at the bench's driver version without version-conditional behaviour that the project's own rules forbid in a logic crate. It cannot be closed from open source. Everything else in this list is a known quantity with a known fix; this one is not yet a known quantity.

9 How to read the code

9.1 Entry points

If you want to understand...	Read
the whole system, fastest	`README.md`, then `ARCHITECTURE.md` — but read §10 of this paper first, because both are stale in specific ways
what a "process" is and why	`crates/kayfabe-core/src/project.rs` — the pure derivation
the source of truth	`crates/kayfabe-core/src/rmgraph.rs` — the refcounted graph and the parked-fact machinery
the transactional apply	`crates/kayfabe-core/src/gpu.rs`, `Gpu::apply` → `Spine::apply`
the forwarding entry points	`crates/kayfabe-fwd/src/lib.rs` — start at `route_doorbell`, then `plan_doorbell`; the ring gate itself now lives one crate over, at `kayfabe-isolate` `VerbPlan::gated_doorbell`, with its compile-fail pin in `crates/kayfabe-isolate/tests/ui/`
the faked GSP	`crates/kayfabe-gsp/src/lib.rs` (the module map and the five seams), then `boot.rs` — `BootPhase`×`QueueState` is the whole boot protocol; the test-side guest is `tests/src/gspworld.rs`
the concurrency design	`crates/kayfabe-rt/src/lock.rs` (ranks) then `device.rs` `verb_op` — the whole plan/execute/commit shape is one function
the `unsafe` surface	`ls crates/kayfabe-linux-raw/src/*_unsafe.rs` — that is the complete list, by construction
what a test is supposed to look like	`tests/tests/l1_mean.rs` (the arbiter) and `tests/tests/miss_taxonomy.rs` (the sharpest single property)

9.2 Conventions

- `*_unsafe.rs` — any file using the `unsafe` keyword is named this way, and the gate greps whole words so writing *about* the rule (`unsafe_code`, `_unsafe.rs`) never trips it.
- `plan_* / commit_*` — the two halves of the verb seam. `plan_` runs under locks and issues nothing; `commit_` runs after re-acquisition and re-validates.
- `route_* / exec_*` — the route/act split (structural rule R4): resolve identity against device-wide state, then act on one process's state.
- Test names carry their provenance: `t12_`, `t13_`, `t14_` are C-bug regressions; `cb*` are the regression matrix rows; `b1_`–`b6_` are security boundary cases; `i1`–`i4` are the isolation properties; the marker *Mutation-gate kill* in a doc comment names the mutant a test exists to kill.

9.3 Which design documents to read, in order

1. `docs/design/testing_doctrine.md` (321 lines) — read this *first*, even before the architecture. It is the shortest path to understanding what the project believes evidence is.
2. `docs/design/core_state_and_consolidation.md` — the reviewed per-crate state and the L1 hand-off contract.
3. `docs/design/core_security_threat_model.md` — the attacker model and the I1–I4 properties.
4. `docs/design/l1_concurrency.md` §12 and `docs/design/l1_os_shell.md` §14 — the contact logs (the two documents run to 10 265 lines between them). **Read these rather than the summaries**; they are where the design was found to be wrong, which is the part a summary cannot carry.
5. `docs/reference/` — measured ground truth, kept out of the design documents so a wrong fact is corrected in one place. Note the version caveat in `rm_semantics_measured.md` §0 before quoting any number from it.

10 Claims corrected while writing this paper

Each of these is a claim that this paper checked against the tree and found stale or wrong. They are listed because the list is more useful than any individual correction: it is a measurement of the drift rate described in §8.9. None is a defect in the code; all are defects in descriptions of the code.

Rows 1–14 are from the first revision, at `0b78102`. **Eleven of the fourteen have been fixed or superseded in the four days since**, and each row says which — that is the loop working, and it is the single most encouraging measurement in this paper. Row 9 is the instructive exception: it was fixed *in one document*, which then recorded four further documents still citing the retracted number and declared them out of scope. Rows 15–20 are from this revision, at `6ce8599`, four days later. **Two of them are defects in this paper's own previous**

revision, which is the point: a whitepaper is a description of code and decays at the same rate as any other. The figures decayed fastest, because a drawn matrix is exactly as ungated as a sentence.

	Claim, and where	What the tree says
1	*"*`handle_doorbell` *is the ONLY caller of* `RmBackend::ring_doorbell`*"* — `ARCHITECTURE.md:112`, `core_completeness_gate.md:119`, `core_state_and_consolidation.md:174`, `fwd/src/lib.rs:15`	**False as stated.** `ring_doorbell` has exactly one call site and it is inside `Worker::execute`, two hops away. `plan_doorbell` had *two* callers: `handle_doorbell` and `SharedDevice::doorbell` — and the L1 shell path, the one a real guest MMIO write takes, goes through the second and **never enters** `handle_doorbell`. The safety property survived; the cardinality claim did not. **Superseded at** `634b253`: the gate moved into `VerbPlan::gated_doorbell`, the sole constructor of a `#[non_exhaustive]` variant, so the property no longer rests on a cardinality claim at all — it rests on E0639. *A correction that the code then made unnecessary is the best outcome on this list.*
2	`kayfabe-abi` is a *"stub (shape only)"* — `README.md`, `ARCHITECTURE.md`, and its own `Cargo.toml` description	**False**, and **FIXED SINCE** (`e6ba19e`). 7 455 lines of source, a working offline generator, 11 generated structs, a 13-method version-dispatch decode surface, 2 272 lines of oracle tests. All three sites now carry struck-through corrections. *Listed here because the correction is the evidence the house style works.*
3	*"the implementor is the L1 shell* (`kayfabe_rt::SharedDevice`)*"* — `ARCHITECTURE.md:38--43`, `kayfabe-vmm/src/lib.rs:816`, `core/src/gpu.rs:1012`	**No** `impl Device for SharedDevice` **exists.** The only two implementors are test types that wrap an `Arc<SharedDevice>` and forward. **Still true at** `6ce8599`, but **the descriptions are FIXED**: both `ARCHITECTURE.md` and `kayfabe-vmm/src/lib.rs` now state the absence in bold rather than implying the shell already does it.
4	*"283 tests"* — `README.md`, `ARCHITECTURE.md`	**515** measured on 2026-07-27; **547** at this revision. **FIXED SINCE** (`e6ba19e`) — and fixed in the right way: both files now say "in the **500s** as of 2026-07-27" rather than carrying a fresh exact number that would rot again in a week. *Deleting a number is a legitimate repair.*
5	*"four standing CI gates"* — `README.md`	**Six jobs; twelve steps in the** `stable` **job alone.** The list of four omitted the VMM-vocabulary gate, the GPA-accessor gate, three unsafe-containment asserts, the KVM reached-count floor, the fuzz-compile job, the slow job and the mutation job — and, since it was written, the generation-name gate. **FIXED SINCE**: the phrase no longer appears in `README.md`.
6	*"TSan (28 threaded tests, 0 races)"* — `README.md`	28 was the first campaign. The four TSan targets contain **68** `#[test]` functions at this revision (65 at the previous one). The "0 races" half is not disputed here — it was not re-run. **FIXED SINCE**: `README.md` now names 28 as "the first campaign" and says it has not been re-run.
7	*"the two measured-slow tests"* — `README.md`	Four to six, depending on how you count; the `g10_*` capped pair joined later. **FIXED SINCE**: the README now says membership "has grown since it was measured" and points at the `skip_slow!` call sites as the authority instead of quoting a count.
8	*"the conformance suite (23 files)"* — `ARCHITECTURE.md`	**29** files in `tests/tests/` at this revision, plus **6** under `crates/*/tests/` (28 + 5 at the previous one). **FIXED SINCE**: the count was removed rather than updated.
9	`core_mutation_gate.md`'s 91% threshold and its *Reproduce* block	Both described the pre-2026-07-27 four-path scope. **FIXED SINCE** (`e6ba19e`): the document now opens with a banner saying so. *But the correction propagated only one hop* — the same document records that the 92.44% and the 91% floor are "still cited as settled" by `README.md`, `ARCHITECTURE.md`, `l1_architecture_summary.md` and `core_state_and_consolidation.md`, and reports them as out of its own edit scope. A correction that names its own unpropagated instances is the honest form; it is still four live instances.
10	`kayfabe-mmu::walker`'s own doc: *"Also here (skeleton this milestone): the GMMU walk algorithm"* — `mmu/src/lib.rs:30`	There is no algorithm: one trait, one enum, 41 lines, zero constructors, zero implementors — **unchanged at** `6ce8599`. **The doc is FIXED SINCE**: the module comment now carries a struck-through correction naming what it actually contains. *The code did not move; the claim did.*

	Claim, and where	What the tree says
11	`kayfabe-gsp`'s doc: RPC decode *"against* `kayfabe-abi`*'s generated message layouts"*	**Was false; is now true.** At the previous pin `kayfabe-gsp` did not depend on `kayfabe-abi` and nothing did. At `6ce8599` the manifest lists it and five of the eight modules use it. *A doc claim that was aspirational and became correct is still a drift instance — nothing gated it either way.*
12	*"*`Isolate/RmBackend/Vmm` *trait-only (mock-implemented)"* — `ARCHITECTURE.md:31,56`	True for `Isolate/RmBackend`. **False for** `Vmm`: `crates/kayfabe-vmm-kvm` is a real KVM adapter with a real memory plane. **FIXED SINCE** — `ARCHITECTURE.md:31` now carries the struck-through original.
13	*"~14 months of quirk capture"* in the C's architecture and ABI documents	Git: 2026-05-24 to 2026-07-24. **Two months, 739 commits.**
14	The framing "7B LLM at 63 tok/s" as a headline C result	63.4 is a Mode-1 A/B endpoint. The audited parity figure is **0.97× (66.0 guest vs 67.8 host)**. The underlying milestone note still records the pre-fix **20 tok/s** and was never updated. All three numbers are real; they are different builds.
15	`kayfabe-gsp` is a **stub** — `README.md:161`, `ARCHITECTURE.md:68`, and its own `Cargo.toml` description, which still opens *"SKELETON — the faked GSP…"*	**False.** 3 550 lines, 8 modules, 24 dedicated tests, an independent 1 133-line guest model, and a composed run inside the arbiter suite. The two markdown rows at least carry an `[unverified]` hedge telling the reader to read `lib.rs` instead; the `Cargo.toml` description carries none. *This is the same shape as row 2, one crate later, four days after row 2 was written.*
16	`mode2_gsp_port_plan.md` §11-O7 marked **RESOLVED**: *"*`GSP_RUN_CPU_SEQUENCER` *is not implemented, do not emit it"*	**Backwards for our bench, and the status has been withdrawn** (`7af23cb`). That is 610's answer. At 580 it is fully implemented, dispatched, and first on the bootup-window allowlist. The whole plan's citations were taken from 610.43.02 while the bench runs 580.159.04; eight claims changed when the matching tree was vendored. See §8.2.
17	The plan's §1.3: *"the guest declares its own geometry"*, presented as the highest-leverage design choice	**610 only.** At 580 `MESSAGE_QUEUE_INIT_ARGUMENTS` has four fields, not nine, and the queue geometry is compile-time. On the bench nothing is negotiated, so the argument the design leaned on does not apply to the machine it will first run on.
18	**121** bare `ogkm:` citations in the crates' doc comments, 96 of them in `kayfabe-gsp`	**Every one is a 610 path with a 610 line number**, and several name files that do not exist at 580 — the whole `msgq` library moved directories between the tags. The version-tagging rule that fixes the class is normative in the plan and **not applied to the code**; the CI grep that would prevent the recurrence is specified and not written.
19	**This paper's own previous revision:** *"*`kayfabe-abi` *and* `kayfabe-gsp` *are disconnected islands"*	**Half false at** `6ce8599`**.** `abi` has a consumer now (`gsp`); `gsp` still has none outside `tests/`. Re-checked against the manifests, not inherited from the caption.
20	**This paper's own previous revision:** Figure 2, *"all 65* `[dependencies]` *edges are shown"*	**The manifests had 66.** The edge `tests` → `rt` was omitted, in a figure whose stated method is "derived from the manifests". A drawn matrix is exactly as ungated as a sentence.

10.1 What this paper could not verify

- **Any claim requiring a build or a run.** No `cargo` command was executed while writing this, at either revision. Test counts, mutation scores, TSan results, timings and the arm64 run are all read from commit messages, workflow files and design documents, and are attributed as such. The 547 figure in particular is a recorded measurement from `e508a8a`, not one taken here.
- **Anything about NVIDIA's 580 driver that this paper did not read itself.** The §8.2 claims about 580-vs-610 are taken from `docs/design/gsp_580_correction_brief.md` and `mode2_gsp_port_plan.md` §14, which state their own limit: only 580.159.04 and 610.43.02 are vendored and were read directly. Seven further driver tags used to narrow the element-layout break interval are **relayed from another agent's probe and were read by nobody in this chain** [unverified]. The predicate the project derived (`major >= 610`) is safe under either reading because the 610 boundary is the directly-verified one; the sentence "580, 590 and 595 are all on the 48-byte side" is not.
- **Whether the 580 resume path is ever taken.** §11-O7a is open, it is the sharpest risk in §8.2, and the evi-

dence that would settle it — the contents of a sequencer buffer — is in closed GSP-RM firmware. Nothing here narrowed it.

- **"TSan, 0 races."** Not re-run; only the target list and the test count in those targets were checked.
- **The relationship between the C's 2026-06-16 bare-metal result and the 2026-07-25 one-process-per-lifetime finding.** The memory note flags this itself: the finding was not cross-checked against the bare-metal box, and warns against assuming the C regressed.
- **Whether Mode 2 ever produced any display output.** No positive evidence was found, only the plan. Absence of evidence.
- **The #57 defect's exact reproduction rate.** "~1 in 6 under race" is quoted from `crate_maturity_map.md`; it was not reproduced here.

A Measurement method

Every number in this paper that is not attributed to a document was produced by one of the following, run against `/workspace/nvkvm-rs` at `6ce8599` and `/workspace/nvidia-gpu-passthrough` at `2899a74`. No compilation, no test execution and no network access was involved. *The measurements were taken from a clean extraction of the pinned commit* (`git archive 6ce8599`), not from a working tree, because the working tree at the time carried unrelated in-flight edits from concurrent work.

Quantity	Command
implementation lines	`find crates -path '*/src/*' -name '*.rs' | xargs wc -l`
test lines	the same over `tests`, `crates/*/tests`, `fuzz`
per-crate lines	the same, per `crates/<name>`
`#[test]` sites	`grep -rc '#[test]'` over workspace members, excluding the detached `crates/kayfabe-abi/gen` and the `trybuild` ui fixtures
`unsafe` relaxations	`grep -rhoE 'unsafe ({|fn |impl |trait )'` `crates/kayfabe-linux-raw/src/*_unsafe.rs | wc -l` — the same expression the CI ratchet uses
dependency edges	every `Cargo.toml` read in full, cross-checked against `Cargo.lock`
commit counts and dates	`git rev-list --count HEAD`, `git log --format=%ci`
document staleness	`git log -1 --format=%ci` per file, compared against the commit that invalidated the claim

The five figures are TikZ, drawn from the measurements above and from a function-by-function read of the call chains they depict; they are vector, not traced from an image. Figure 2 in particular is generated from the manifest edge list rather than from any design document, which is why it disagrees with the hexagon picture about which crates are "ports".

Two method changes at this revision, both prompted by the figures being where the drift was worst. Figure 3 now names *symbols* rather than `file:line` pins: six of its pins had drifted in fourteen commits, and the repository had already made the same change to its own 105 pins in `3fd1455`. And Figure 2's edge list was re-derived from scratch rather than edited, which is how the omitted `tests` → `rt` edge was found (§10 row 20).

One closing note on how to use this paper. If you are here to find what is wrong with kayfabe, the highest-yield reading order is: §8.2 (the open resume handoff, which has no proposed resolution), §8.7 (the packaging cost the floor decision did not buy its way out of), §7's final subsection (what the suite cannot prove), the two contact logs named in §9.3, and then the code — starting with `VerbPlan::gated_doorbell` and `Spine::apply`, which between them contain most of the load-bearing decisions. Do not start with `README.md`; §10 lists six things wrong with it (one of them since fixed), and it will cost you an hour.